GENTLE ALGEBRAS AND ITS CONNECTIONS TO SURFACE TRIANGULATIONS AND BRAUER GRAPH ALGEBRAS

BY

DERRICKSON YMBON

DEPARTMENT OF MATHEMATICS

NORTH-EASTERN HILL UNIVERSITY

SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF

MASTER OF SCIENCE IN MATHEMATICS

To

NORTH-EASTERN HILL UNIVERSITY

SHILLONG – 793022

JULY 2026

GENTLE ALGEBRAS AND ITS CONNECTIONS TO SURFACE TRIANGULATIONS AND BRAUER GRAPH ALGEBRAS

A dissertation submitted in the partial fulfilment

of the requirement for the degree of

Master of Science

in

MATHEMATICS

by

DERRICKSON YMBON

(Roll No. 24MScMath19)

Batch: 2024–2026

Under the supervision of

DR. ARDELINE MARY BUHPHANG

PROF. THOMAS BRÜSTLE

Department of Mathematics

North-Eastern Hill University

Shillong – 793022, India

July 2026

DEDICATED

TO

MY MUM AND AUNT

NORTH-EASTERN HILL UNIVERSITY

July, 2026

DECLARATION

I, **Derrickson Ymbon**, hereby declare that the dissertation entitled **"Gentle Algebras and its Connections to Surface Triangulations and Brauer Graph Algebras"** is the outcome of original work carried out by me under the supervision of Dr. Ardeline Mary Buhphang and Prof. Thomas Brüstle, in the Department of Mathematics, School of Physical Sciences, North-Eastern Hill University, Shillong, for the degree of **Master of Science** in **Mathematics**. To the best of my knowledge, the contents of this dissertation are not excerpts of any previous work carried out by me or anybody else. I have not submitted this dissertation or any part of it for any degree in any other university or institution.

I am submitting this dissertation to North-Eastern Hill University, Shillong, for the degree of Master of Science in Mathematics.

Signature of the Candidate

Derrickson Ymbon
Batch: 2024–2026
Roll No.: 24MScMath19

Countersigned by:

Dr. A. M. Buhphang
Head, Department of Mathematics,
NEHU, Shillong

Dr. Ardeline Mary Buhphang
Supervisor

Prof. Thomas Brüstle
Supervisor

CERTIFICATE

I certify that the dissertation entitled **"Gentle Algebras and its Connections to Surface Triangulations and Brauer Graph Algebras"** submitted by **Derrickson Ymbon** to North-Eastern Hill University, Shillong, for the award of the degree of **Master of Science in Mathematics** is a bonafide record of academic work carried out by him under our supervision. I certify that the sources from which ideas have been borrowed have been duly referred to. The content of this dissertation has not been presented for the award of any degree anywhere before.

This dissertation may be placed before the examiners for evaluation and necessary formalities. We certify that this dissertation is worthy of consideration by the examiners.

Signature of the Supervisor

Dr. Ardeline Mary Buhphang
Associate Professor
Department of Mathematics
North-Eastern Hill University
Shillong, INDIA

Place: Shillong, India

Date: ______________

Signature of the Supervisor

Prof. Thomas Brüstle
Professor
Department of Mathematics
Université de Sherbrooke
Sherbrooke, CANADA

Place: Sherbrooke, Canada

Date: ______________

ACKNOWLEDGEMENT

The resilience that God has given me throughout the entirety of this academic session has been nothing but mind-blowing. So, for that I thank God and his soldiers who are none other but my fellow colleagues, my classmates, Mr. Rishabh Goswami, and most importantly, my advisors Dr. Ardeline Mary Buhphang and Prof. Thomas Brüstle.

Ma'am Ardeline has been a godsend for me during this tenure, and without her dedication and Rishabh's, I would have never expected to complete my thesis in this manner. The hard work they put in to make this project come to life is indispensable, and I owe a lifetime to them. I would also like to thank the Mathematics Department at NEHU, Shillong for making my academic journey throughout the entire 2 years extremely memorable. The help that the faculty have provided is incomparable, dare I say, to a lot of universities, and the tools and teachings they have provided will remain with me throughout my lifetime. My fellow colleagues Reality, Kerlan, and Yoowanka have been an incredibly fun group to work with and for that, I am grateful.

My aunt and her children mean the world to me, and her everlasting support throughout my stumble-filled journey has been invaluable, whose help I cannot put into words honestly, so thank you.

Of course, this all would not have possible without my friends and my classmates at NEHU, from the Mathematics Department as well as those who are not from the Mathematics Department.

The character growth throughout the entirety of these 2 years has been immense, and for that I am grateful to every single one of the above mentioned. So, thank you.

PREFACE

In the first chapter of this dissertation, we present and define preliminary concepts relating to algebras and modules, and consequently quivers and quiver algebras and their representations. We also state the trichotomy of the representation types of finite-dimensional associative algebras, and we take a closer look at the tame type, more specifically, special biserial algebras by an example.

In the second chapter, we define a subclass of special biserial algebras, called the gentle algebras, and we show that the algebras that are obtained from marked oriented surfaces are gentle, which means that we have a connection between the two objects. We also look at certain results and propositions, for example the trivial extension of gentle algebras being symmetric special biserial. We also look at a collection of information called ribbon data and special types of graphs called ribbon graphs, and we display an explicit example of how we can construct surfaces using ribbon graphs

The third chapter introduces Brauer Graphs and Brauer Graph Algebras, which are algebras that are obtained by an algorithm from said combinatorial data that is Brauer Graphs. We also look at the the connection between gentle algebras and Brauer Graph algebras and the ways we can obtain one out of the other.

Finally, in the fourth chapter of this thesis, we document an important relation between algebras that arise out of triangulations with the same parameters. More specifically, a relation between their derived categories, and the derived categories of their trivial extensions. The potential research interests and scope are also discussed, in particular the geometric models of the module categories of gentle algebras as per Baur,Coelho-Simoes, and the geometric models of the derived categories of gentle algebras as presented by Opper,Plamondon,Schroll.

Chapter 5 is the appendix. It is where the preliminaries for the 4th chapter are presented, including category theory, in particular homotopy category, derived category and derived equivalences. There is also a short exposition on localisation of categories included in the appendix.

Contents

List of Abbreviations x

1 Introduction and Preliminaries 1

1.1 Introduction . . . 1

1.2 Preliminaries . . . 2

1.2.1 Modules and Algebras . . . 2

1.2.2 Bound Quiver Algebras and their Representations . . . 7

1.2.3 Representation Types of Algebras: The Trichotomy and Gabriel's Theorem . . . 12

1.2.4 Special Biserial Algebras . . . 14

2 Gentle Algebras and Surface Triangulations 16

2.1 Gentle Algebras . . . 16

2.2 Ribbon Graph . . . 18

2.3 Construction of Ribbon Graphs from Gentle Algebras: Ribbonning 20

2.4 Surface Triangulations and Gentle Algebras . . . 26

3 Brauer Graph Algebras 32

3.0.1 Brauer Graphs . . . 32

3.0.2 Quivers from Brauer Graphs . . . 35

3.0.3 Set of relations and definition of Brauer graph algebras . . . 39

3.0.4 Indecomposable Projective modules over Brauer Graph Algebras . . . 43

3.0.5 Gentle Algebras and Brauer Graphs: The connection . . . 45

3.0.6 Slices . . . 48

4 Future Scope: Derived Equivalence and Geometric Models of Gentle Algebras **50**

4.1 Derived Equivalence of Gentle Algebras 51

4.2 Geometric Models of the Module Category of Gentle Algebras . . . 54

4.3 Geometric Models of the Derived Category of Gentle Algebras . . 55

5 Appendix **57**

5.1 Homotopy Category 57

5.2 Localisation of Categories 59

5.3 Derived Category 61

5.4 Derived Equivalence 63

Bibliography **65**

List of Abbreviations

e_i	ith idempotent
Q	Quiver
K	Algebraically closed field
A	Finite-dimensional associative algebra
M	A–module
$\operatorname{rad} M$	Radical of M
KQ	Path algebra of a quiver Q
KQ/I	Path algebra of a bound quiver (Q, I)
B	Brauer Graph
KQ_B/I_B	Brauer Graph Algebra
C_v	Special cycle at a vertex v
Γ	Ribbon data
S	Topological Surface
$\mathscr{A}$	Abelian Category
$\operatorname{Hom}(A, B)$	Class of Morphisms
$\mathbf{Ch}(R)$	Chain complex of R–modules
$\mathcal{K}(R)$	Homotopy Category of R–modules
$\mathcal{D}(\mathscr{A})$	Derived category of $\mathscr{A}$
$\mathcal{D}^b(\mathscr{A})$	Bounded derived category of $\mathscr{A}$

Chapter 1

Introduction and Preliminaries

1.1 Introduction

The representation theory of finite-dimensional algebras has been a very active area in the world of mathematics as of late. More importantly, there is a trichotomy in the theory, where the representation type of algebras can be of three types namely finite, tame or wild, thanks to the work of Drozd in [4]. Algebras of finite type are most easily understood. An algebra is said to be of finite type if the number of isoclasses of indecomposable modules is finite. Algebras of tame representation type, on the other hand, have an infinite number of isoclasses of indecomposable modules, but our understanding of them is more-or-less "controlled", in the sense that all but a finite number of these indecomposable modules can be parametrized by 1-parameter families, and wild representation type is the least understood among the three.

There is an important class of algebras called the *special biserial* algebras that was shown to be tame [14, 15]. This class contains some other classes of algebras that are crucial in representation theory, but here we will mainly be interested in two, namely the gentle algebras and Brauer graph algebras [11].
Gentle algebras and Brauer graph algebras are closely related; that is, we can obtain Brauer graph algebras using the trivial extension of the gentle algebras, and we can go back to the original algebra via the concept of admissible cuts [13].

This relationship also has a geometric aspect. Gentle algebras arise naturally from triangulations of marked oriented surfaces whose marked points lie on the boundary [2]. Given such a triangulation, one can associate a quiver with relations, and the corresponding *bound quiver algebra* is gentle. Thus, gentle algebras provide a bridge between the representation theory of finite-dimensional algebras and the combinatorics and geometry of marked surfaces.

Ribbon graphs provide another geometric model relevant to this picture. A ribbon graph encodes a cyclic ordering of half-edges at each vertex, and this cyclic ordering allows the graph to be realized on an oriented surface. Brauer graphs may be viewed as ribbon graphs equipped with additional multiplicity data. Consequently, the study of gentle algebras, Brauer graph algebras, ribbon graphs, and triangulated surfaces reveals a network of connections between algebra, combinatorics, and geometry.

The aim of this chapter is to document the algebraic and geometric background needed for these connections. We begin with classical algebraic structures such as associative algebras and quivers, path algebras, and bound quiver algebras. We then recall the notions of finite, tame, and wild representation type, including Drozd's theorem and Gabriel's theorem. After that, we introduce special biserial algebras, discuss trivial extensions and Brauer graph algebras, and finally explain the role of ribbon graphs and triangulated surfaces in the construction of gentle algebras.

1.2 Preliminaries

1.2.1 Modules and Algebras

We begin by recalling some preliminary notions on modules and algebras that will be used throughout the chapter. The definitions and the notations used herein are standard; we mainly follow Assem–Simson–Skowroski [1, Chapter I].

Definition 1.2.1. Let R be a unital ring. A left R-module is a set M equipped with a binary operation $+ : M \times M \to M$ such that $(M, +)$ is an abelian group, and an action of the ring R on M (i.e., a map $R \times M \to M$ denoted by $r \cdot m, \forall r \in R,\ m \in M$) such that $\forall r, s \in R$ and $m, n \in M$, the following properties are satisfied:

(M1) $(r+s) \cdot m = r \cdot m + s \cdot m$

(M2) $r \cdot (m+n) = r \cdot m + r \cdot n$

(M3) $(r \cdot s) \cdot m = r \cdot (s \cdot m)$

(M4) $1 \cdot m = m$

Right R-modules can be defined analogously by putting the elements of R on the right. If the ring is commutative, then the notion of left and right modules are equivalent.

We will later see that modules over path algebras are one of the main objects of study in representation theory of finite-dimensional algebras.

Some example of modules are:

Example 1.2.2.

1. The ring R can be seen as an R-module with the action being just the ordinary multiplication inside the ring itself.
2. If I is a left(right) ideal of the ring, then I is a left(right) module over R
3. If G is an abelian group, then G is a $\mathbb{Z}$-module

Definition 1.2.3. Let M be an R-module. A subset $N \subseteq M$ is called a submodule of M if N is an abelian subgroup of M and $rn \in N$ for all $r \in R$ and $n \in N$.

Definition 1.2.4. A Λ-module P is said to be *projective* if for every surjective Λ-module homomorphism

$$g : M \twoheadrightarrow N$$

and every Λ-module homomorphism

$$f : P \to N,$$

there exists a Λ-module homomorphism

$$h : P \to M$$

such that

$$g \circ h = f.$$

Equivalently, every homomorphism from P to a quotient module lifts through the quotient map.

Definition 1.2.5. A Λ-module I is said to be *injective* if for every injective Λ-module homomorphism

$$g : M \hookrightarrow N$$

and every Λ-module homomorphism

$$f : M \to I,$$

there exists a Λ-module homomorphism

$$h : N \to I$$

such that

$$h \circ g = f.$$

Equivalently, every homomorphism into I extends along an inclusion.

Definition 1.2.6. An R-module M is called simple if $M \neq 0$ and its only submodules are 0 and M. It is called indecomposable if $M \neq 0$ and M cannot be written as a direct sum of two nonzero submodules.

Definition 1.2.7. If M is an R-module, then the radical of M, denoted by $\operatorname{rad} M$, is defined to be the intersection of all maximal submodules of M. That is,

$$\operatorname{rad} M = \bigcap_{N \text{ maximal submodule of } M} N.$$

Definition 1.2.8. Let P be an indecomposable projective Λ-module. The *heart* of P is defined to be the quotient

$$\operatorname{Ht}(P) = \frac{\operatorname{rad}(P)}{\operatorname{soc}(P)},$$

where $\operatorname{rad}(P)$ denotes the radical of P and $\operatorname{soc}(P)$ denotes the socle of P.

We now recall the notion of an associative algebra over a field K, also referred to as a K-algebra.

Definition 1.2.9. Let K be an algebraically closed field. A K-algebra is a unital ring A together with a K-vector space structure such that the scalar multiplication is compatible with the ring multiplication, That is,

$$\lambda(ab) = (\lambda a)b = (a\lambda)b = a(\lambda b) = (ab)\lambda \quad \text{for all } \lambda \in K \quad \text{and } a, b \in A$$

Example 1.2.10.
- The polynomial ring $K[x]$ in the indeterminate x is a K-algebra. In fact it is an infinite dimensional K-algebra.
- The ring $M_n(K)$ of all square matrices of order n with entries from K is an n^2-dimensional K-algebra

Definition 1.2.11. A' is said to be a sub-algebra of A if A' is a K-vector subspace of A and $a_1a_2 \in A'$ $\forall a_1,\ a_2 \in A'$, $1 \in A'$. A left(right) **ideal** of an algebra A is a K-vector subspace of A such that for all $a \in A, ai$ (resp. ia) $\in I$, for all $i \in I$.

If I is both a left and right ideal, then we call I a *two sided ideal*

Now we define the radical of an algebra, which is an important object.

Definition 1.2.12. Let K be an algebraically closed field. The radical of a K-algebra A, also called the **Jacobson radical**, denoted by rad A is defined as the intersection of all maximal right ideals of A. That is, if $\mathcal{M}$ is the set of all maximal right ideals in A, then

$$\operatorname{rad} A = \bigcap_{I \in \mathcal{M}} I$$

Example 1.2.13.
1. If we consider a field K as an algebra over itself, then $\operatorname{rad} K$ is simply the zero ideal. That is $\operatorname{rad} K$=0
2. Let A be the K-algebra $K[x]/(x^s)$. Then the Jacobson radical of A is given by the ideal $(x + (x^s))$

Definition 1.2.14. Let A be an K-algebra. Then we have the following definitions:

1. An element $e \in A$ is called an **idempotent** element if $e^2 = e$.
2. An element e is said to be a **central** idempotent if it is an element of the center of A,i.e. $ea = ae \forall a \in A$
3. Two idempotents e_1 and e_2 are called **orthogonal** if $e_1 e_2 = e_2 e_1 = 0$.
4. An idempotent is said to be **primitive** if e cannot be decomposed into two non-zero orthogonal idempotents. That is, if $e = e_1 + e_2$ for some orthogonal elements e_1, e_2, then either $e_1 = 0$ or $e_2 = 0$

Definition 1.2.15. A $K-$algebra A is said to be **connected** if the only central idempotents are 0 and 1, or equivalently A is connected if it is indecomposable as a direct product of two nonzero algebras: $A \ncong A_1 \times A_2$

Definition 1.2.16. A finite-dimensional K-algebra A is called **basic** if

$$A/\operatorname{rad} A \cong K \times K \times \cdots \times K$$

as K-algebras.

Equivalently, A is basic if, as a semisimple algebra, $A/\operatorname{rad} A$ is a direct product of copies of K. Basic algebras are important because every finite-dimensional algebra is Morita equivalent to a basic algebra; see [1, Chapter I, Section I.6].

For a finite-dimensional K-algebra A, we write

$$D(A) = \operatorname{Hom}_K(A, K)$$

for the K-dual of A.

Definition 1.2.17. An algebra A is said to be symmetric if $A \cong D(A)$ as A-A-bimodules.

It is important to note that while A and $D(A)$ are isomorphic as finite-dimensional K-vector spaces, the isomorphism in the above definition is of bimodules, where the multiplication is defined as follows:

$$(af)(x) = f(xa) \quad \text{and} \quad (fb)(x) = f(bx)$$

for $a, b \in A$ and $f \in D(A)$. There exists an equivalent formulation of this. For that we recall that a linear form f from a $K-$algebra A to a field K is said to be symmetric if $f(ab) = f(ba)$. Then an algebra is symmetric if there exists a symmetric linear form $f : A \to K$ such that $\ker f$ does not contain any non-zero left ideal in the algebra.

1.2.2 Bound Quiver Algebras and their Representations

In this subsection we will recall the notions of quivers, path algebras, bound quiver algebras, and their representations. The primary reference for the material is Assem–Simson–Skowroski [1, Chapters II and III].

Definition 1.2.18. Let K be an algebraically closed field. A quiver is a quadruple $Q = (Q_0, Q_1, s, t)$, where

- Q_0 is the set of all vertices
- Q_1 is the set of all arrows
- A map $s : Q_1 \to Q_0$ which maps an arrow $\alpha \in Q_1$ to its starting point $s(\alpha)$
- A map $t : Q_1 \to Q_0$, which maps the same arrow to its ending point $t(\alpha)$.

In other words, if α is an arrow, then $s(\alpha)$ is the "source" of the arrow and $t(\alpha)$ is the "target" of the arrow. So, a quiver is essentially just a directed graph where loops and parallel arrows are allowed. As such, when we take a loop α, then the source and target of the loop is the same. For example, take the following quiver

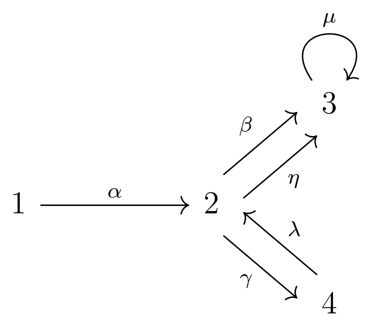


Figure 1: Quiver $Q^{(1)}$

The above diagram is a quiver where $Q_0 = \{1, 2, 3, 4\}, Q_1 = \{\alpha, \beta, \gamma, \lambda, \mu, \eta\}$. where $s(\alpha) = 1$ and $t(\alpha) = 2$,$s(\beta) = 2 = s(\eta)$ and $t(\beta) = 3 = t(\eta)$, $s(\mu) = t(\mu) = 3$,

$s(\lambda) = 4$ and $t(\lambda) = 2$, and $s(\gamma) = 2$ and $t(\gamma) = 4$

Definitions 1.

1. A **path** in Q from a vertex i to another vertex j is a sequence of arrows $c = (i \mid \alpha_1, \alpha_2, ...\alpha_m \mid j)$, where the target of the previous arrow is the source of the next arrow.

2. The **trivial** path e_v at a vertex i is the path $i||i$. That is, it is the path which stays at i

For every quiver Q, we can also get its corresponding *path algebra* KQ, which is an algebra over the field K, and whose underlying vector space is generated by all the paths in the quiver. In other words, the basis of KQ is the set of all paths in Q. Also important to note that if the quiver has a loop or an oriented cycle, then there would be infinite paths, and thus the path algebra would be infinite dimensional. The path algebra for the quiver $Q^{(1)}$ in Figure 1 is infinite dimensional whereas, the path algebra of a finite acyclic quiver is finite dimensional.

Like any other algebra (or vector space), the multiplication can be determined by the multiplication of two basis elements. So, here if c and c' are any two basis elements of KQ, then the multiplication $c \cdot c'$ is defined by

$$c \cdot c' = \begin{cases} cc', & \text{if } t(c) = s(c') \\ 0, & \text{otherwise} \end{cases}$$

Here cc' is the concatenation of two paths in Q (note that concatenation of two paths is only possible if the ending vertex of c is the starting vertex of c', so that means multiplication in KQ is possible only if concatenation in Q is possible. If it is not we simply define it to be 0). The trivial paths act as idempotents, and if Q_0 is finite, then identity of KQ namely, $1_{KQ} = \sum_{i \in Q_0} e_i$.

Let us denote by $\mathcal{R}$ the two-sided ideal of KQ generated by all the arrows in Q. Another two sided ideal I is called **admissible** if there exists a positive integer

$m \geq 2$ such that

$$\mathcal{R}^m \subseteq I \subseteq \mathcal{R}^2$$

Here, $\mathcal{R}^m$ is simply the decomposition

$$\mathcal{R}^m = \bigoplus_{l \geq m} KQ_l$$

, where KQ_i is the subspace of KQ generated by the paths of length i, for $i = m, m+1, \cdots$ and as a vector space, $\mathcal{R}^m$ has a basis being the set of all paths of length greater than or equal to m. So, if I is an admissible ideal, then it would contain all the paths greater than or equal to m, which would mean that the algebra obtained by factoring out I from KQ, that is, the algebra KQ/I is finite dimensional. We also say that the algebra KQ/I is given by a quiver and relations. Also, the condition $I \subset \mathcal{R}^2$ guarantees that when we quotient I out from the path algebra, then we would not cut out any arrows.

The reason for introducing bound quiver algebras is that they provide a way of studying finite-dimensional algebras through quivers and relations. More precisely, from the point of view of representation theory, one may first replace a finite-dimensional algebra by a Morita equivalent basic algebra. The following result then allows us to realize such an algebra as a quotient of a path algebra by an admissible ideal.

Theorem 1.2.19 ([1, Chapter II, Section II.3]). *Let A be a basic connected finite-dimensional K-algebra. Then there exists a finite connected quiver Q_A, called the ordinary quiver of A, and an admissible ideal I of KQ_A such that*

$$A \cong KQ_A/I.$$

Consequently, every connected finite-dimensional K-algebra is Morita equivalent to a bound quiver algebra.

Thus, for representation-theoretic purposes, the study of finite dimensional algebras may often be reduced to the study of algebras of the form KQ/I. The quiver Q encodes the combinatorial data, while the admissible ideal I records the relation among paths. We now recall how modules over such algebras can be described as representations of quivers with relations.

Definition 1.2.20. A **representation** of a quiver Q consists of a family

$$M = (M_i, \varphi_\alpha)_{i \in Q_0, \alpha \in Q_1},$$

where to every vertex $i \in Q_0$ we assign a K-vector space M_i, and to every arrow $\alpha \in Q_1$ we assign a K-linear map

$$\varphi_\alpha : M_{s(\alpha)} \longrightarrow M_{t(\alpha)}.$$

A representation M is called finite-dimensional if each M_i is finite-dimensional.

Example 1.2.21. Consider the quiver

$$1 \xrightarrow{\alpha} 2.$$

A representation $M = (M_i, \varphi_\alpha)$ of Q consists of two K-vector spaces M_1 and M_2, together with a single K-linear map

$$\varphi_\alpha : M_1 \to M_2.$$

For instance, taking

$$M_1 = K, \qquad M_2 = K, \qquad \varphi_\alpha = \mathrm{id}_K$$

gives the representation

$$K \xrightarrow{1} K.$$

More generally, one could take $M_1 = K^m$, $M_2 = K^n$, and let φ_α be any K-linear map.

Definition 1.2.22. Let $M = (M_i, \varphi_\alpha)$ and $N = (N_i, \psi_\alpha)$ be two representations of a quiver Q. A morphism $f : M \to N$ is a family of K-linear maps

$$f_i : M_i \to N_i, \qquad i \in Q_0,$$

such that for every arrow $\alpha : i \to j$, the diagram

$$\begin{tikzcd} M_i \arrow[r, "\varphi_\alpha"] \arrow[d, "f_i"'] & M_j \arrow[d, "f_j"] \\ N_i \arrow[r, "\psi_\alpha"'] & N_j \end{tikzcd}$$

commutes. Equivalently,

$$f_j \varphi_\alpha = \psi_\alpha f_i.$$

If $p = \alpha_1 \alpha_2 \cdots \alpha_m$ is a path in Q, then a representation M assigns to p the composition

$$\varphi_p = \varphi_{\alpha_m} \cdots \varphi_{\alpha_2} \varphi_{\alpha_1}.$$

Now let (Q, I) be a bound quiver. A representation of (Q, I) is a representation $M = (M_i, \varphi_\alpha)$ of Q such that all relations in I are satisfied. More precisely, if

$$\rho = \sum_{r=1}^{n} \lambda_r p_r \in I,$$

where each p_r is a path in Q, the paths p_r have the same source and target, and $\lambda_r \in K$, then we require

$$\sum_{r=1}^{n} \lambda_r \varphi_{p_r} = 0.$$

In this case, we say that M is bound by I, or that M satisfies the relations in I.

We denote by $\mathrm{Rep}_K(Q, I)$ the category of representations of Q satisfying the relations in I, and $\mathrm{rep}_K(Q, I)$ the category of representations of Q satisfying the relations in I, and similarly let $\mathrm{Mod}\, A$ denote the category of right A−modules, where $A = KQ/I$ and $\mathrm{mod}\, A$ denote the category of finitely generated right A−modules

By Theorem III.1.6 of Assem–Simson–Skowroski, if $A = KQ/I$, where Q is a finite connected quiver and I is an admissible ideal of KQ, then there is a K-linear equivalence

$$\mathrm{Mod}\, A \simeq \mathrm{Rep}_K(Q, I),$$

which restricts to an equivalence

$$\mathrm{mod}\, A \simeq \mathrm{rep}_K(Q, I).$$

Thus, when $A = KQ/I$ is fixed, we shall identify finite-dimensional A-modules with finite-dimensional representations of the bound quiver (Q, I); see [1, Theorem III.1.6].

In particular, since every finite-dimensional K-algebra is Morita equivalent to a basic algebra, and every basic connected finite-dimensional K-algebra can be realised as a bound quiver algebra KQ/I, the representation theory of such algebras may be studied through representations of bound quivers.

1.2.3 Representation Types of Algebras: The Trichotomy and Gabriel's Theorem

We now recall the representation types of finite-dimensional algebras. As mentioned in the introduction, there is a trichotomy in the representation theory of finite-dimensional algebras, where we have three representation types, namely finite, tame and wild.

Definition 1.2.23.

1. An algebra is said to be of **finite representation type** if there are only finitely many isoclasses of indecomposable modules over the algebra.

2. An algebra is said to be of **tame representation type** if there are infinitely many isoclasses of indecomposable modules, but all but a finite number of these isoclasses can be parametrized by 1-parameter families.

3. An algebra is said to be of **wild representation type** if it contains the representation theory of the free algebra $k\langle x, y\rangle$

In other words, representation-finite type algebras are the easiest to understand, tame algebras are less understood than finite and wild algebras are the least understood among the three.

The following theorem, due to Drozd, gives the fundamental division of finite-dimensional algebras.

Theorem 1.2.24 (Drozd's theorem [4])**.** *Let A be a finite-dimensional algebra over an algebraically closed field K. Then A is tame or wild.*

Equivalently, if one separates the representation-finite case from the tame representation-infinite case, then every finite-dimensional algebra is of finite, tame, or wild representation type.

We now look at the case arising from path algebras of acyclic quivers. Consider the quiver

$$1 \xrightarrow{\alpha} 2$$

Figure 2: Quiver $Q^{(2)}$

The path algebra $KQ^{(2)}$ is of finite representation type. In fact, up to isomorphism, its indecomposable representations are

$$K \to 0, \qquad 0 \to K, \qquad K \xrightarrow{1} K.$$

Thus there are only three indecomposable representations.

If we add one more arrow from 1 to 2 to $Q^{(2)}$, we obtain the Kronecker quiver

$$1 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 2.$$

The corresponding path algebra for this quiver is no longer of finite representation type. In fact, the Kronecker algebra is a standard example of a **tame representation type algebra**.

We now recall Gabriel's celebrated theorem, which classifies the connected acyclic quivers whose path algebras are of finite representation type.

Theorem 1.2.25 ([10, Theorem 3.1])**.** *Let Q be a connected acyclic quiver. Then the path algebra KQ is of finite representation type if and only if the underlying graph of Q, obtained by forgetting the orientation of the arrows, is a Dynkin diagram of type $\mathbb{A}, \mathbb{D}, \mathbb{E}$.*

In particular, the quiver $1 \to 2$ has underlying graph $\mathbb{A}_2$, and hence its path algebra is of finite representation type by Gabriel's theorem. The Kronecker quiver, on the other hand, has two parallel arrows and its underlying graph is not a Dynkin diagram of type $\mathbb{A}$, $\mathbb{D}$, or $\mathbb{E}$. Therefore, its path algebra is not of finite representation type.

1.2.4 Special Biserial Algebras

We now consider an important class of bound quiver algebras obtained by imposing restrictions both on the quiver Q and on the admissible ideal I. The first restriction is the number of arrows going in and out of a vertex in Q. Another condition is a condition on the ideal where multiple continuations and pre-continuations are killed in the algebra. Formally stated, we have the two conditions:

(G1) For every vertex $v \in Q_0$, at most two arrows start at v and at most two arrows end at v.

(G2) For every arrow β in Q, there exists at most one arrow α such that $\alpha\beta \notin I$ and there exists at most one arrow γ such that $\beta\gamma \notin I$.

By placing the conditions above on the quiver, we obtain a special biserial algebra.

Definition 1.2.26. A finite-dimensional K-algebra A is called *special biserial* if there exists a quiver Q and an admissible ideal I of KQ such that A is Morita equivalent to KQ/I and KQ/I satisfies conditions (G1) and (G2).

The conditions (G1) and (G2) have a simple combinatorial interpretation. For (G2), we can phrase it as if for every arrow β there exist multiple arrows, say $\alpha_1, \alpha_2, \alpha_3$ before β in Q, then at most one of those continuations with β will not be in the ideal, i.e., say $\alpha_1\beta \notin I$, then $\alpha_2\beta \in I$ and $\alpha_3\beta \in I$, and the same goes for arrows after β. In other words, the paths $\alpha_2\beta$ and $\alpha_3\beta$ will not survive in the algebra. The condition (G1) on the other hand, controls the number of arrows entering or leaving each vertex. It means that for every vertex $v \in Q_0$, at most two arrows start at v and at most two arrows end at v.

Special biserial algebras form a significant class of tame algebras. The following result is due to Wald and Waschbüsch.

Theorem 1.2.27 ([15])**.** *Special biserial algebras are of tame representation type.*

The proof of this theorem is based on a classification of the indecomposable modules over a special biserial algebra. In general, the indecomposable modules over special biserial algebras are classified using string and band modules, but that is beyond the scope of this work, and will not be discussed here.

Example 1.2.28. Consider the Kronecker quiver

$$1 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 2.$$

Let $I = 0$. Then $KQ/I = KQ$ satisfies condition (G1), since there are exactly two arrows starting at vertex 1, no arrows starting at vertex 2, no arrows ending at vertex 1, and two arrows ending at vertex 2. It also satisfies condition (G2), since there are no composable paths of length two. Hence the Kronecker algebra (KQ) is special biserial.

For each $\lambda \in K$, we have a representation

$$K \underset{\lambda\,\mathrm{id}_K}{\overset{\mathrm{id}_K}{\rightrightarrows}} K.$$

These representations already indicate the presence of one-parameter families, which is characteristic of tame representation type. In fact, the Kronecker algebra is a standard example of a tame hereditary algebra.

Chapter 2

Gentle Algebras and Surface Triangulations

In this chapter, we take a look at surfaces, in particular surfaces with marked points, or punctures. We define triangulations of such surfaces, and we explore the types of algebras that are obtained out of these triangulations. This chapter closely follows the articles [2, 11]

2.1 Gentle Algebras

In addition to the conditions (G1) and (G2) imposed on the quiver Q, we can add more conditions which results in another class of algebras:

Definition 2.1.1. A finite dimensional algebra Λ is called a **gentle** algebra if it is Morita equivalent to the bound quiver algebra KQ/I where Q and I satisfies (G1) and (G2) along with the following conditions.

The conditions are:

(G3) For every arrow β in Q, there exists at most one arrow α such that $\alpha\beta \in I$ and at most one arrow γ such that $\beta\gamma \in I$.

(G4) The ideal I is generated by paths of length 2.

In other words, if the quiver Q and the ideal I in KQ/I satisfies conditions (G1),(G2),(G3) and (G4), then the algebra is gentle.

So we see that the class of special biserial algebras contains the class of gentle algebras, since we can obtain gentle algebras by imposing more conditions on the quiver Q and I.

Let us take a look at one example of a gentle algebra. Consider the following quiver:

$$
\begin{tikzcd}
 & 2 \arrow[dr, "\gamma"] & & \\
1 \arrow[ur, "\beta"] & & 3 \arrow[r, "\eta"] \arrow[dl, "\delta"] & 5 \\
 & 4 \arrow[ul, "\alpha"] & &
\end{tikzcd}
$$

where $I = \langle \alpha\beta, \gamma\delta \rangle$. This ideal is admissible by taking $n = 4$ for $\mathcal{R}^n$. (G1) is satisfied from the diagram since there are at most two arrows going in and out of all the vertices.

Now we check for (G2). For β, we can see that no arrow β' exists such that $\beta\beta' \notin I$. Similarly, we can check the same for all the arrows and it can be shown that (G3) holds for every arrow as well. (G4) is clearly true as well since paths of length 2 generate the ideal I

Now we define a collection of data Γ' called **Ribbon Data**. First, it will be convenient to define a set of *half-edges*:

Definition 2.1.2. Without loss of generality, define a half-edge incident on a vertex $v \in E$ to be an edge e_+ such that the edge e_+ and the other half edge e_- incident on v' make up the entire edge e connecting v and v'. Here v and v' may not necessarily be distinct

Now the ribbon data Γ' is a collection of data as follows:

- V is a finite collection of vertices.
- E is a collection of half edges.
- $s : E \to V$ is a map which sends each half edge to the vertex it is incident on.
- $i : E \to E$ is an involution with no fixed points.

Intuitively, i can be thought of as a map which sends a half-edge e incident on a vertex $v \in V$ to the other half edge e' incident on v' which when glued together, e and e' make up the same edge connecting v and v'.

2.2 Ribbon Graph

Definition 2.2.1. Ribbon data that is equipped with a permutation $\sigma : E \to E$ that corresponds to the sets $s^{-1}(v)$ is called a ribbon graph, or a fat graph. To be more precise, a ribbon graph is ribbon data along with the permutation σ, or $\Gamma = (V, E, s, i, \sigma)$

The name *fat* is due to the fact that the edges of the ribbon graph can literally be fattened to make a surface.

For example, take the following ribbon graph

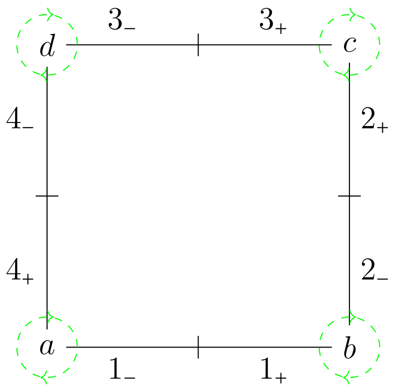


Figure 1: Ribbon Graph Γ

Here, we have the half edges to be $\{1_+, 1_-, 2_+, 2_-, 3_+, 3_-, 4_+, 4_-\}$. As we can see from Definition 2.1.2, the half edges 1_+ and 1_- make up the entire edge connecting a and b and likewise for the other edges.

Since s is a function which sends the half edges to the vertices they are adjacent to, then $s(4_+) = s(1_-) = d$, $s(1_+) = s(2_-) = b$, $s(3_+) = s(2_+) = c$ and $s(4_-) = s(3_-) = d$. Then we have the involution $i(1_+) = 1_-$. This is an involution because if we apply i again, then we get $i(i(1_+) = i(1_-)$. But $i(1_-) = 1+$, by definition of i. So, $i(i(1_+) = 1_+$. Similarly, we get $i(2_-) = 2_+$, $i(3_-) = 3_+$, and $i(4_-) = 4_+$. The cyclic ordering around a is given by $s^{-1}(a) = \{1_-, 4_+\}$. So $\sigma_a = (1_- \ 4_+)$. The arguments are similar for the other vertices.

Definition 2.2.2. [7] Let Γ be a connected ribbon graph, and let us denote the valency of v by n if $\mathrm{val}(v) = n_v$ The ribbon surface $S(\Gamma)$ is constructed by gluing polygons as follows:

1. For every vertex with $\mathrm{val}(v) \geq 1$, let P_v be an oriented $2n_v$−polygon.

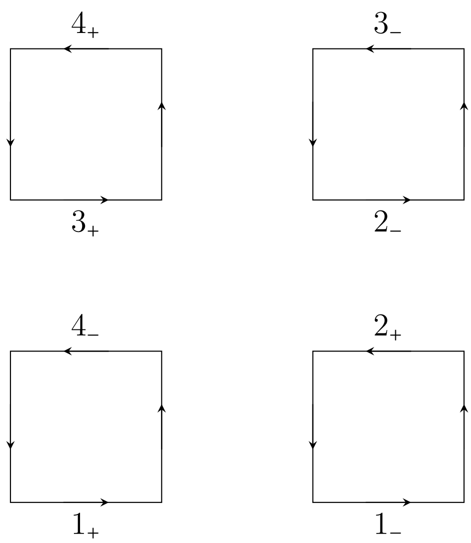
2. Following the cyclic orientation, label every other side of P_v with the half-edges $e \in E$ such that $s(e) = v$.

3. For any half-edge e of Γ, identify the side of P_v labelled e with the side of the polygon $P_{s(i(e))}$ labelled $i(e)$, respecting the orientations of the polygons.

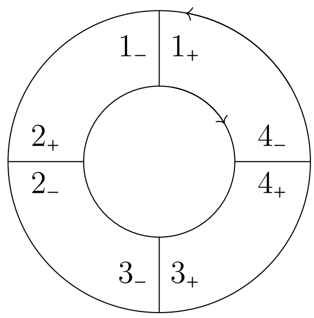
The third step is just fancy terminology for just taking the side of one polygon labelled e_+ as in step 1, and then gluing that side to $i(e_-)$ which is e_+, while of course preserving the orientation of the polygon.

Now let us take Γ and use definition 2.2.2 to make the desired polygon.

First, in step 1 we note that for vertex a, we get a 4−gon, or a square, since val(a)=2. The same goes to all the vertices. Then we label every other side of the square with the set $s^{-1}(a)$, which is $\{4_-, 1_+\}$ here. Continuing in this process, we will get 4 squares like as follows



So here, we do step 3, which is just gluing the sides of the polygons which make up the entire edge in Γ. That means gluing the side 4_- with 4_+ and so on to get the following surface, which is an annulus.



Ribbon graphs are useful precisely because they have a geometric interpretation as above.

They are also important because from gentle algebras we can construct ribbon graphs. We discuss this method here.

2.3 Construction of Ribbon Graphs from Gentle Algebras: Ribbonning

Suppose we have a gentle algebra $A = KQ/I$. Define a maximal path in a quiver to be a path which cannot be further extended on both sides without becoming a relation. That is,

Definition 2.3.1. A path p in a bound quiver (Q, I) is said to be maximal if there is no $\alpha, \beta \in Q_1$ with $\alpha p, p\beta \notin I$

Let $\mathcal{S}$ be the set of all maximal paths in KQ/I (or (Q, I)), and let $\mathcal{S}'$ denote the set of all trivial paths e_v such that

1. v is a source or a sink with exactly one arrow starting or ending at v respectively.

2. There exist unique arrows α and β such that $s(\alpha) = v$ and $t(\beta) = v$ and $\alpha\beta \notin I$

Let $\mathcal{S} \cup \mathcal{S}' = \overline{\mathcal{S}}$. Define the associated ribbon graph $\Gamma_A = (\Gamma_0, \Gamma_1, s, i, \sigma)$ to be the ribbon graph whose set of vertices is precisely $\mathcal{S} \cup \overline{\mathcal{S}}$. That is, $\Gamma_0 = \overline{S}$

For the set of half-edges Γ_1, let us take an element $p = i_1 \xrightarrow{\alpha_1} i_2 \xrightarrow{\alpha_2} \cdots \xrightarrow{\alpha_t} i_t \in \bar{\mathcal{S}}$. Denote the vertex in Γ_A corresponding to this path by ν_p. Then there is an edge E_{i_k} between ν_p and another vertex $\nu_{m_{i_k}}$ if and only if the paths p and m_{i_k} pass through the same vertex i_k in the path p.

For the ordering or permutation, let $E_1, E_2, \cdots, E_t$ be the edges corresponding to the vertices in the path p as considered above. Then the ordering is formed by setting $E_1 < E_2 < \cdots E_t$ and adding the edge E_1 to this linear ordering completes it to a cyclic ordering, and thus we have completed our ribbonning.

Example 2.3.2. For example, take the gentle algebra A given by the following quiver

$$\begin{tikzcd}
 & & 3 \arrow[dr, "\alpha_3"] & & \\
1 \arrow[r, "\alpha_1"] & 2 \arrow[ur, "\alpha_2"] \arrow[dr, "\alpha_6"'] & & 4 \arrow[r, "\alpha_4"] & 6 \\
 & & 5 \arrow[ur, "\alpha_5"'] & &
\end{tikzcd}$$

Figure 1: An example of a gentle algebra

Here, $I = \langle \alpha_1\alpha_2, \alpha_5\alpha_4 \rangle$ We have the extended set of maximal paths $\overline{\mathcal{S}} = \{p_1 = \alpha_1\alpha_6\alpha_5, p_2 = \alpha_2\alpha_3\alpha_4, e_1, e_3, e_6, e_5\}$. The set of vertices $\Gamma_0 = \overline{\mathcal{S}}$. $\alpha_1\alpha_6\alpha_5$ and $\alpha_2\alpha_3\alpha_4$ share a vertex, so there will be an edge between them. Similarly for e_1 and $\alpha_1\alpha_6\alpha_5$, e_3 and $\alpha_2\alpha_3\alpha_4$, e_6 and $\alpha_2\alpha_3\alpha_4$, and e_5 and $\alpha_1\alpha_6\alpha_5$. So we get the following graph

$$\begin{tikzcd}[arrows=-]
e_1 \arrow[r, "E_1"] & p_1 \arrow[r, "E_5"'] \arrow[rrr, bend left, "E_2"] \arrow[rrr, bend right, "E_4"'] & e_5 & e_6 \arrow[r, "E_6"] & p_2 \arrow[r, "E_3"] & e_3
\end{tikzcd}$$

We have the path $1 \xrightarrow{\alpha_1} 2 \xrightarrow{\alpha_6} 5 \xrightarrow{\alpha_5} 4$. Therefore, we have the cyclic ordering $E_1 < E_2 < E_5 < E_4 < E_1$, and for the path $2 \xrightarrow{\alpha_2} 3 \xrightarrow{\alpha_3} 4 \xrightarrow{\alpha_5} 6$, we have the cyclic ordering around the path to be $E_2 < E_3 < E_4 < E_6 < E_2$. In section 3, subsection 3.0.5, we see that the quiver $\overline{Q}$, which is obtained from the simply completing the maximal path to a cyclic path, is the quiver obtained in 3.0.5

Here we state and prove a crucial theorem in representation theory which presents a relation between gentle algebras and special biserial algebras, and specifically later we will see, a relation between gentle algebras and Brauer Graph Algebras.

First we recall that if A is a finite-dimensional algebra, then its trivial extension $T(A)$ is defined as

$$T(A) = A \ltimes D(A)$$

where $D(A) = \mathrm{Hom}_K(A, K)$ has the structure of an A, A-bimodule where K is an algebraically closed field. Here the underlying vector space is $A \oplus D(A)$, where the multiplication is defined as

$$(a, f)(b, g) = (ab, fb + ag)$$

Proposition 2.3.3. *The trivial extension $T(A)$ of an algebra A is symmetric.*

Combined with the condition that if A is gentle, we get the following property which is a strong characterisation of the trivial extensions of gentle algebras. First, we state a lemma.

Lemma 2.3.4. *[13] Let $\Lambda = kQ/I_\Lambda$ and $\Delta = kQ/I_\Delta$ be symmetric special biserial algebras with common underlying quiver Q. Suppose that, for every vertex of Q, the indecomposable projective modules of Λ and Δ have k-bases given by the same paths in Q. Then Λ and Δ are isomorphic as algebras.*

Theorem 2.3.5. *Let A be a gentle algebra. Then the trivial extension of $A, T(A)$ is symmetric special biserial.*

Proof. First, we note that the from the definition of the trivial extension $T(A)$, there is a short exact sequence of $T(A)$–bimodules

$$0 \to D(A) \xrightarrow{f} T(A) \xrightarrow{g} A \to 0$$

,since Im f = Ker g, and the map $f(x) = (x, 0)$ is injective as a $T(A)$–bimodule homomorphism.

For every vertex i in $Q_{T(A)}$, let $e_i \in T(A)$ be the corresponding primitive idempotent. Then another short exact sequence of right $T(A)$-bimodules is induced if we multitply each object in the sequence by e_i

$$0 \to e_i D(A) \to e_i T(A) \to e_i A \to 0$$

where $e_i D(A)$ is the indecomposable injective A-module at i, $e_i A$ is the indecomposable projective A–module at i.

Let us take two maximal paths m_i and n_i which pass through some vertex i of the quiver Q. That is, m_i and n_i are of the form

$$m_i = i_1 \xrightarrow{\alpha_1} i_2 \xrightarrow{\alpha_2} \cdots \xrightarrow{\alpha_{k-1}} i_{k-1} \xrightarrow{\alpha_k} i_k = i \xrightarrow{\alpha_{k+1}} \cdots \xrightarrow{\alpha_m} i_m$$

$$n_i = j_0 \xrightarrow{\beta_1} j_1 \xrightarrow{\beta_2} \cdots \xrightarrow{\beta_{t-1}} j_{t-1} \xrightarrow{\beta_t} j_t = i \xrightarrow{\beta_{t+1}} \cdots \xrightarrow{\beta_n} j_n$$

The indecomposable projective right module $e_i A$ has a basis consisting of all the paths starting at i. By condition (G1), we have that there are at most two arrows starting at i and j, namely α_k and β_t, provided that the arrows exist, for some vertex j. By condition (G2), there is at most one arrow α_{k+1} and at most one arrow β_{t+1} such that $\alpha\alpha_{k+1}$ is in the ideal and $\beta_t\beta_{t+1}$. Again by the same condition there exists at most one arrow α_{k+2} and β_{t+2} such that $\alpha_{k+1}\alpha_{k+2}$ and $\beta_{k+1}\beta_{k+2}$ is not in the ideal. Continue this process until we reach α_m for the α_i's and β_n for the β_i's. Then since m_i and n_i are maximal, this continuation must stop at one arrow for each chain. Call those arrows γ and δ. That is $\alpha_m\gamma$ is zero and $\beta_n\delta$ is zero. Therefore the paths $m_i' = i \xrightarrow{\alpha_{k+1}} \cdots \xrightarrow{\alpha_m} i_m$ and $n_i' = i \xrightarrow{\beta_{t+1}} \cdots \xrightarrow{\beta_n} j_n$ are the two unique non-zero paths starting at i, if they exist of course. The following figure 2 helps in visualising the paths

$$\begin{tikzcd}
i_1 \arrow[r, "\alpha_1"] & i_2 \arrow[r, "\alpha_2"] & \cdots \arrow[r, "\beta_{k-1}"] & i_{k-1} \arrow[dr, "\alpha_k"] & & i_{k+1} \arrow[r] & \cdots \arrow[r, "\alpha_m"] & i_m \\
& & & & i \arrow[ur, "\alpha_{k+1}"] \arrow[dr, "\beta_{t+1}"] & & & \\
j_1 \arrow[r, "\beta_1"] & j_2 \arrow[r, "\beta_2"] & \cdots \arrow[r, "\beta_{k-1}"] & j_{t-1} \arrow[ur, "\beta_t"] & & j_{t+1} \arrow[r] & \cdots \arrow[r, "\beta_n"'] & j_n
\end{tikzcd}$$

Figure 2: The paths m_i and n_i

This means that the string module $M(s)$ as in the following figure is the corresponding module to e_iA, since every indecomposable module over a gentle algebra is either a string module or a band module.

$$\begin{tikzcd}
i_1 \arrow[r, "\alpha_1"] & i_2 \arrow[r, "\alpha_2"] & \cdots \arrow[r, "\beta_{k-1}"] & i_{k-1} \arrow[dr, "\alpha_k"] & & i_{k+1} \arrow[r, rightsquigarrow] & \cdots \arrow[r, rightsquigarrow, "\alpha_m"] & i_m \\
& & & & i \arrow[ur, rightsquigarrow, "\alpha_{k+1}"] \arrow[dr, rightsquigarrow, "\beta_{t+1}"] & & & \\
j_1 \arrow[r, "\beta_1"] & j_2 \arrow[r, "\beta_2"] & \cdots \arrow[r, "\beta_{k-1}"] & j_{t-1} \arrow[ur, "\beta_t"] & & j_{t+1} \arrow[r, rightsquigarrow] & \cdots \arrow[r, rightsquigarrow, "\beta_n"'] & j_n
\end{tikzcd}$$

Figure 3: The string s indicated by the squiggly lines

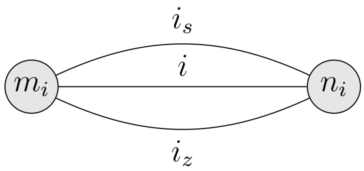
In the extended quiver $\overline{Q}$, there are arrows $i_m \xrightarrow{\alpha_{m_i}} i_1$ and $j_n \xrightarrow{\beta_{m_i}} j_1$

and so, the projective-injective (recall that projectives and injectives coincide in symmetric algebras) indecomposable $T(A)$-module $e_iT(A)$ is given by the following diagram

$$\begin{tikzcd}
 & i \arrow[dl] \arrow[dr] & \\
i_{k+1} \arrow[d] & & j_{k-1} \arrow[d] \\
\cdots \arrow[d] & & \cdots \arrow[d] \\
i_m \arrow[d] & & j_m \arrow[d] \\
i_1 \arrow[d] & & j_1 \arrow[d] \\
i_2 \arrow[d] & & j_2 \arrow[d] \\
\cdots \arrow[d] & & \cdots \arrow[d] \\
i_{k-1} \arrow[dr] & & j_{k-1} \arrow[dl] \\
 & i &
\end{tikzcd}$$

where the heart of the module rad $e_iTA)$/soc $e_iT(A)$ is given by the direct sum of the uniserial modules, say U_1 and U_2 generated by the direct string $i_{k+1} \xrightarrow{\alpha_{k+1}} \cdots \xrightarrow{\alpha_{m-1}} i_m \xrightarrow{\beta_{m_i}} i_1 \xrightarrow{\alpha_0} i_2 \xrightarrow{\alpha_1} \cdots \xrightarrow{\alpha_{k-2}} i_{k-1}$, and $j_{l+1} \xrightarrow{\gamma_{l+1}} \cdots \xrightarrow{\gamma_{n-1}} j_n \xrightarrow{\beta_{n_i}} j_0 \xrightarrow{\gamma_0} j_1 \xrightarrow{\beta_1} \cdots \xrightarrow{\beta_{l-2}} j_{l-1}$ respectively. Note that the above diagram is in the extended quiver $\overline{Q}$ of $T(A)$

Now that we have the two maximal paths m_i and n_i, we have two vertices of the ribbon graph Γ_A of A. Denote these two vertices by ν_{m_i} and ν_{n_i} Since both of the paths pass through i, therefore there will be an edge E_i connecting the two vertices. Suppose i_s and i_z are two other vertices which the paths m_i and n_i pass through. Then we have the following ribbon graph Γ'_A



Let A' denote the Brauer Graph Algebra obtained from the graph Γ'_A. Then the indecomposable projective module $P_i(A')$ clearly has the same path basis in the algebra $K\overline{Q}/I = T(A)$ as well as A'. Thus $T(A)$ and A' are isomorphic, by lemma 2.3.4.

Now there is the case where either n_i or m_i correspond to the trivial path at i. Assume without loss of generality that $n_i = e_i$. Then $e_iT(A)$ is uniserial corresponding to the string $i \xrightarrow{\alpha_{k+1}} \cdots \xrightarrow{\alpha_m} i_m \xrightarrow{\alpha_{m+1}} i_1 \xrightarrow{\alpha_1} i_2 \xrightarrow{\alpha_2} \cdots \xrightarrow{\alpha_{k-1}} i_{k-1} \xrightarrow{\alpha_k} i$, the edge E_i in Γ_A corresponding to i is a loner vertex(refer to chapter 3). So, similarly from the case when neither of the maximal paths were trivial paths, we can see that the indecomposable projective $P_i(A)$ has the same path basis in the path algebras of Q and $\overline{Q}$ in this case as well. Thus by the lemma 2.3.4, the isomorphism is proved. □

2.4 Surface Triangulations and Gentle Algebras

In this section, we look at triangulations of a marked oriented surface whose marked points all lie on the boundary, and we explore the class of algebras that are obtained from these triangulations.

From here on, when we say surface, we mean a clockwise-oriented surface S with a finite set of marked points M, denoted by the pair (S, M) and we only look at the surfaces where the marked points all lie on the boundary,i.e. $M \subset \partial S$. The pair (S, M) is referred to as an unpunctured bordered surface with marked points. We can think of M as points that are equally spaced on the boundary that make S into a polygon, but in general that is not the case.

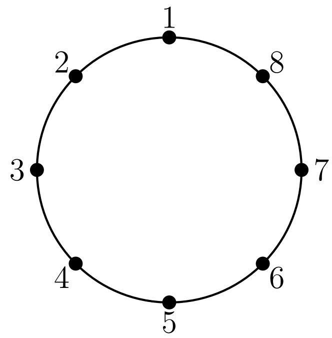


Figure 1: A surface S with marked points $\{1, 2, 3, 4, 5, 6, 7, 8\}$

First, we recall that a curve in a topological space X is a continuous map $\alpha : [0, 1] \to X$. Now we define an *arc* in (S, M)

Definition 2.4.1. An arc in the surface (S, M) is a curve η in S with the following properties :

(ST1) The endpoints of η are marked points in M

(ST2) η does not intersect itself, except possibly at its endpoints.

(ST3) The intersection of η and ∂S is non-empty only at the endpoints of S

(ST4) η does not cut out a monogon

There will be two main types of arcs considered in this study: *boundary arcs* and *internal arcs*

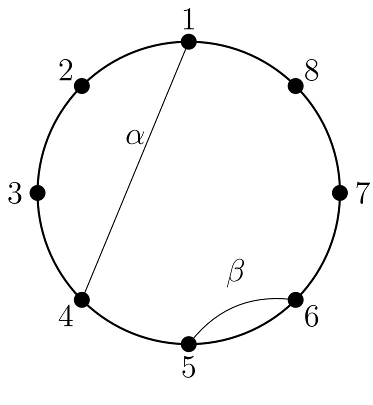


Figure 2: An internal arc and a boundary arc

An arc η is called a boundary arc if it is homotopic to a curve δ in the boundary of S that intersects M only at the endpoints.

An arc η' is called an internal arc if it is not a boundary arc, that is, it is not homotopic to a curve that is in the boundary of S with endpoints only at M. All arcs in the surface are considered up to homotopy. In Fig 2, the arc α is an internal arc and β is a boundary arc, since it is homotopic to another curve in the boundary that intersects the boundary only at the marked points 5 and 6.

Proposition 2.4.2. *The number of internal arcs of an n-gon is $n-3$*

Now we move on to *triangulations* of a surface.

Definition 2.4.3. A triangulation is a maximal set of non-crossing internal arcs in the surface S

The following figure is a triangulation

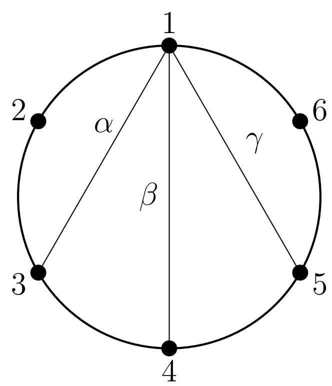


Figure 3: A triangulation of an hexagon

While the following is not

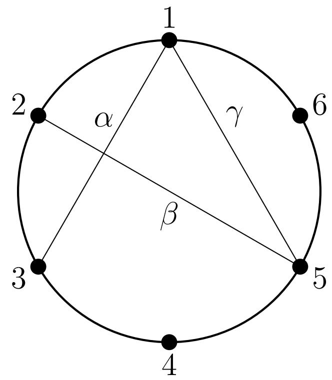


Figure 4: A “non-triangulation"

If T is a triangulation of a surface S, we can obtain a quiver Q_T, where the vertices of the quiver are given by the internal arcs, and the arrows are given by the following process:

Let us consider a triangulation of an octagon, or an 8-gon.

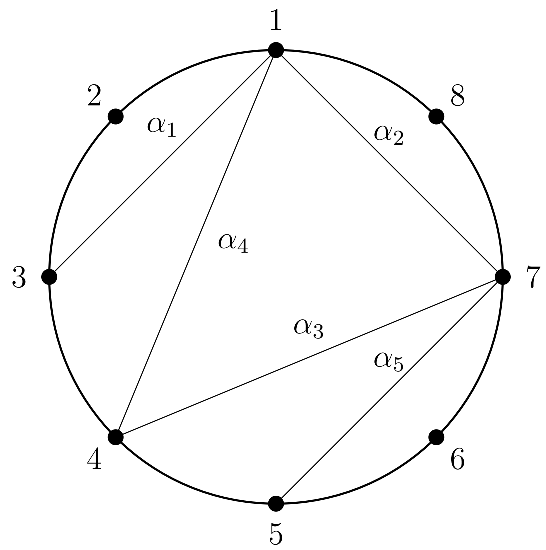


Figure 5: A triangulation of an octagon

Let us consider a triangulation of an octagon, or an 8-gon as in Fig 5. Then we look at an internal triangle, that is, a triangle formed by at least 2 internal arcs, and we take two internal arcs and look at the common vertex of those two arcs. Here, let us take the triangle formed by α_1, α_4, and the boundary $\overline{34}$. Since the surface is oriented clockwise, we have a clockwise orientation around the vertex 1, and therefore we have the arrow from the vertex $\alpha_4 \xrightarrow{a} \alpha_1$. Similarly, we have the arrow b from α_4 to α_3, the arrow c from α_5 to α_3, the arrow d from α_5 to α_3 and the arrow e from α_3 to α_2. Then we have the following quiver

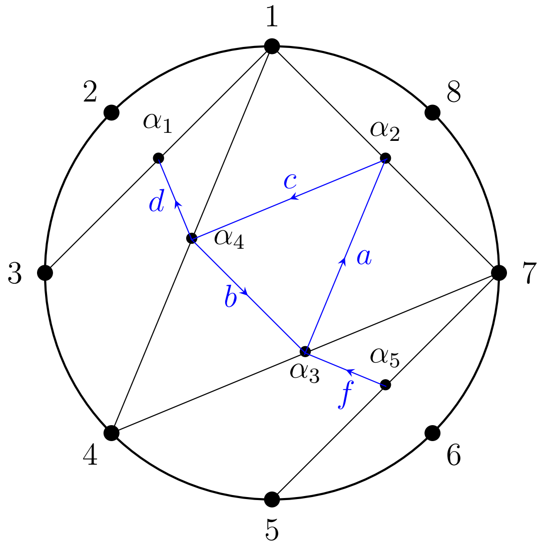


Figure 6: The quiver Q_T

It can be seen that there are counter-clockwise oriented triangles in the internal triangles of the triangulation, and we get the quiver Q_T, with the arrows being $\{a, b, c, d, e, f\}$ and the vertices being $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5\}$

In each internal triangle Δ of a triangulation T, there is an oriented cycle $a_\Delta b_\Delta c_\Delta$ in the quiver Q_T. The ideal I_T is generated by the arrows in the triangle of the form $a_\Delta b_\Delta, b_\Delta c_\Delta$ and $c_\Delta a_\Delta$. That is, $I_T = \langle a_\Delta b_\Delta, b_\Delta c_\Delta, c_\Delta a_\Delta \rangle$.

The quiver obtained from the triangulation, and consequently the path algebra KQ_T/I_T satisfies some nice properties, which we discuss in this section.

Before we do that, first we list a few definitions

Definition 2.4.4. 1. A *boundary triangle* in a triangulation T is a triangle such that two of its sides lie in the boundary of S

2. Let $\delta_1, \delta_2, \cdots \delta_n$ be the boundary components in S, and let m_i be the number of marked points on δ_i, for $i = 1, 2 \cdots n$. Let d_i be the number of boundary triangles incident with δ_i, and let g be the genus of S Then the sequence of numbers

$$g, n, (m_1, d_1), (m_2, d_2) \cdots (m_n, d_n)$$

are called the parameters of T.

In Fig 5, the parameters of the triangulation are

$$0, 1, (8, 3)$$

Since the surface is homeomorphically a disc with no punctures or holes in the interior, it has only one boundary component. The number of boundary triangles incident with the single boundary component, is 3. They are the triangles $\overline{123}$, $\overline{187}$, and $\overline{567}$

Proposition 2.4.5. *[2] The algebra $A_T = KQ_T/I_T$ is a gentle algebra.*

Chapter 3

Brauer Graph Algebras

Brauer graph algebras are finite-dimensional symmetric algebras defined from combinatorial data. A Brauer graph consists of a finite graph together with a cyclic ordering of the edges around each vertex and a multiplicity function. From this data, one constructs a quiver with relations, and hence an associated algebra.

In this chapter, we recall the construction of Brauer graph algebras, following Schroll's survey [12]. We also discuss their connection with gentle algebras through trivial extensions and admissible cuts. Alongside the standard definitions and results, we include several explicit examples to illustrate the construction of the associated quivers, special cycles, and defining relations.

3.0.1 Brauer Graphs

Definition 3.0.1. A Brauer Graph is a tuple $B = (B_0, B_1, m, o)$, where

1. (B_0, B_1) is a finite connected unoriented graph with vertex set B_0 and edge set B_1

2. m, called the multiplicity, is a function which assigns to each vertex a positive integer.

3. o is called the **orientation** of G, and is given as follows : for each vertex $v \in B_0$, there is a cyclic ordering of the edges incident with v. For example, this cyclic ordering may be clockwise or anti-clockwise, and we denote this cyclic ordering by the $<$ sign. If v is incident to only one edge e and $m(v) = 1$,

then the cyclic ordering is simply given by e. If v is adjacent to atleast one vertex or $m(v) > 1$, then the cyclic ordering at v is given by $e < e$

In this survey, we refer to the graph (B_0, B_1) as simply B for convenience of notation.

So, if we denote the set of half edges by E, then $|B_1| = 2|E|$.

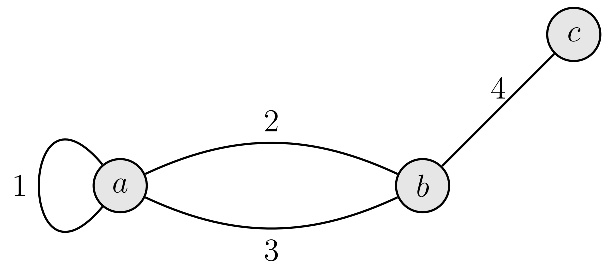
Now we will be able to talk about the *valency* of a vertex, denoted by $\text{val}(v)$.

Definition 3.0.2. The valency of a vertex $v \in B_0$ is the number of half-edges incident on v

In other words, $\text{val}(v)$ is simply the degree of v in terms of half-edges.

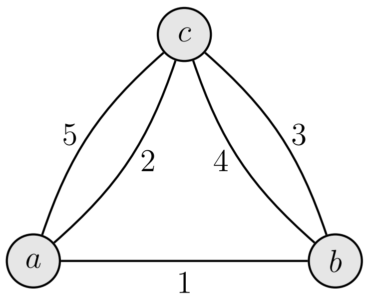
Remark 3.0.3. As we can see, while most of the vertices in the Brauer Graphs considered here are such that $m(v)\text{val}(v) > 1$, there may exist some vertex v' such that $m(v)\text{val}(v) = 1$. This means that either the multiplicity is one or the valency is one, and since all of the vertices in our Brauer Graphs is one, this means that $\text{val}(v') = 1$, and we will call the only edge incident on that vertex v' a *truncated* edge, and we call the vertex v a *lone* vertex, or simply, a *loner*.

Example 3.0.4. 1) Consider the following Brauer graph $B^{(1)} = (B_0^{(1)}, B_1^{(1)}, m_1, o_1)$



Here, we set $m(i) = 1$ for all vertices $i = a, b, c$. Here, we can take the cyclic ordering as the clockwise ordering, and thus the ordering of the edges around the vertex a is given by $1 < 1 < 2 < 3 < 1$, around b is given by $2 < 3 < 4 < 2$ and for c is just 1, since $m(c) = 1$. The number of half-edges incident on a is 4, so $\text{val}(a) = 4$. Counting for b, we get that $\text{val}(b) = 3$, and finally, $\text{val}(c) = 1$, since there is only one edge incident on c. Here, we see that edge 4 is a truncated edge since $m(c)\mathrm{val}(c) = 1$, and vertex c is a lone vertex/loner.

2) Consider the following Brauer graph $B^{(2)} = (B_1^{(2)}, B_0^{(2)}, m_2, o_2)$

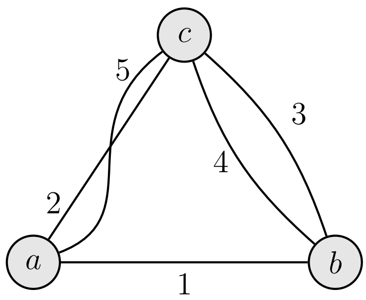


We again set $m(a) = m(b) = m(c) = 1$ for convenience. So here, around vertex a, we take the clockwise ordering, which would give us the ordering $1 < 2 < 5 < 1$, for b, we have us $1 < 4 < 3 < 1$, and for c, we have $2 < 3 < 4 < 5 < 2$.

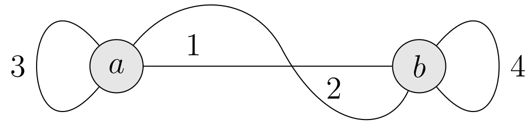
Here, $\mathrm{val}(a) = \mathrm{val}(b) = 3$, and $\mathrm{val}(c) = 4$

3) Let us take the following graph $B^{(3)} = (B_0, B_1, m_3, o_3)$. Here the cyclic ordering around vertex a is $1 < 2 < 5 < 1$, vertex b is $3 < 1 < 4 < 3$, and at vertex c is $5 < 3 < 4 < 2 < 5$ due to the nature of the edge 5.

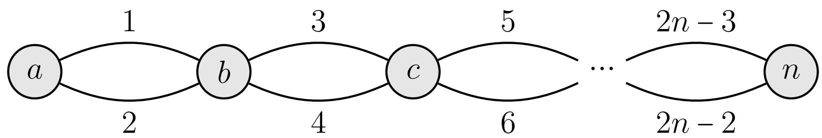
Again the valencies of both the vertices are equal to 3



4) Let us consider the following graph $B^{(4)} = (B_0^{(4)}, B_1^{(4)}, m_4, o_4)$ Again, we take the cyclic ordering in the clockwise sense. Then the ordering around a is $1 < 3 < 3 < 2 < 1$, and the ordering around b is $1 < 2 < 4 < 4 < 1$. The valencies are both equal, that is $\mathrm{val}(a) = \mathrm{val}(b) = 4$



5) Let us consider a more general example of a graph of the following form with n vertices and $2(n-1)$ edges. Then the valencies of all the vertices



except a and n are 4 each The cyclic ordering around vertex a is $1 < 2$, around b is $1 < 3 < 2 < 4$, around c is $3 < 5 < 6 < 4$ and for vertex $n-1$ is $2n-5 < 2n-3 < 2n-2 < 2n-4$. In the same manner, call the above graph $B^{(5)} = (B_0^{(5)}, B_1^{(5)}, m_5, o_5)$

3.0.2 Quivers from Brauer Graphs

In this section, we will be explaining the method on how to construct quivers from Brauer graphs, which is very powerful on the study of Brauer graph algebras.

Let a be a vertex of some Brauer graph B, and let $i_1, i_2, ..., i_n$ be the edges in B incident on a. Here, the i_j's are not necessarily distinct, that is, we might have $i_j = i_k$, for some j, k if the corresponding edge is a loop. Now suppose that the cyclic ordering of these edges around a is given by $i_1 < i_2 < ... < i_n < i_1$. Then we call $i_1, i_2, ..., i_n$ the *successor* sequence at a, and we call i_{j+1} the successor of i_j, for $1 \leq j \leq n-1$ and i_1 is the successor of i_n. Now i_{k-1} is the *predecessor* of i_k, for $2 \leq k \leq n$, and i_n is the predecessor of i_1.

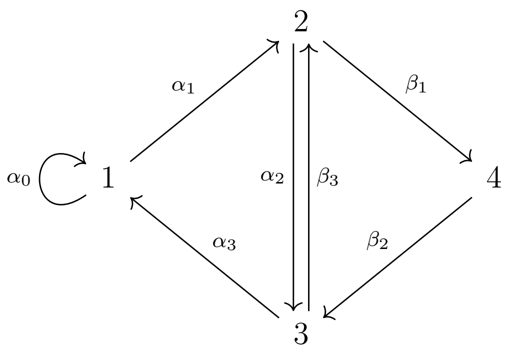
Here, we note that if $\text{val}(v) = 1$ and $m(v) > 1$, then the cyclic ordering at v is $e < e$ for some edge i, so i is both the predecessor and successor of itself. But if $m(v) = 1$, then the edge i does not have a predecessor or successor.

Now, given a Brauer graph $B = (B_0, B_1, m, o)$, we define a quiver $Q_B = (Q_0, Q_1, s, t)$ in the following way: The set of vertices Q_0 is given by the set of edges B_1 in B, where we denote the vertex i in Q_0 by the same edge i in B_1. In other words, if i is an edge in B, then i is a vertex in Q. The arrows in Q are induced by the orientation o in B. More precisely, let i and j be two edges in B with a common vertex v, such that j is a direct successor of i in the cyclic ordering of v. Then we have an arrow $\alpha : i \to j$ in Q_B, and we say that α is given by the successor relation (i, j). Since every arrow in Q_B starts at an edge in B as well as ends at an edge in B, therefore there are at most two arrows starting and ending at every vertex of Q_B, since there cannot be more than two orientations around a specific vertex $v \in V$. Every vertex $v \in V$ such that $m(v)\mathrm{val}(v) \geq 2$, gives rise to an oriented cycle in C_v in Q_B, since if $\mathrm{val}(v) = 1$, then $m(v) > 1$, and we must have a cycle $i < i$ in B, for some edge i, or a loop in Q_G. If $m(v) = 1$, then $\mathrm{val}(v) > 1$, then we will have the following ordering in G at the least : $i < j < i$, which will give us an oriented cycle in Q_G. These cycles C_v are unique up to cyclic permutation. Then, if we take C_v from its cyclic permutation class such that $s(C_v) = i = t(C_v)$, then we call $C_v^{m(v)}$ the *special* i-cycle at v.

For $i \in Q_0$, and $v \in V$, a special i-cycle at v is not necessarily unique. However, there are at most two special i-cycles at v for any $i \in B_1$, and $v \in V$, and this happens exactly when i is a loop. An example of this is in Example 3.0.4 (1)

Example 3.0.5. In this example, we obtain the quivers from the Brauer graphs in examples 3.0.4, and we will also list a few of the special cycles.

1)



Special cycles in Q_{G_1} are the special 1-cycles at a corresponding to $\alpha_0\alpha_1\alpha_2\alpha_3$ and $\alpha_1\alpha_2\alpha_3\alpha_0$.

The special 2-cycle at a is given by $\alpha_2\alpha_3\alpha_0\alpha_1$.

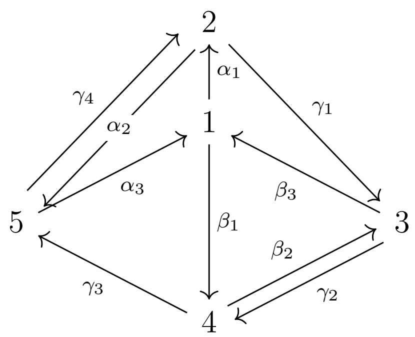
The special 3-cycle at a is given by $\alpha_3\alpha_0\alpha_1\alpha_2$.

The special 2-cycle at b is given by $\beta_1\beta_2\beta_3$.

The special 4-cycle at b is given by $\beta_2\beta_3\beta_1$.

The special 3-cycle at b is $\beta_3\beta_1\beta_2$

2)



The special 1-cycle at a is given by $\alpha_1\alpha_2\alpha_3$.

The special 2-cycle at a is given by $\alpha_2\alpha_3\alpha_1$.

The special 5-cycle at a is given by $\alpha_3\alpha_1\alpha_2$.

The special 1-cycle at b is given by $\beta_1\beta_2\beta_3$.

The special 4-cycle at b is given by $\beta_2\beta_3\beta_1$.

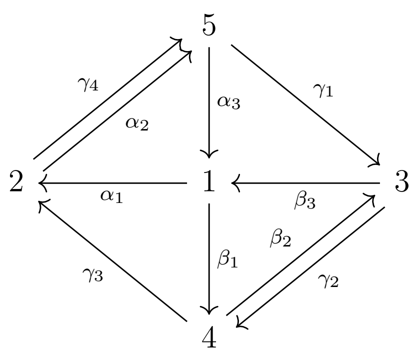
The special 3-cycle at b is given by $\beta_3\beta_2\beta_1$.

The special 2-cycle at c is given by $\gamma_1\gamma_2\gamma_3\gamma_4$

The special 3-cycle at c is given by $\gamma_2\gamma_3\gamma_4\gamma_1$

The special 4-cycle at c is given by $\gamma_3\gamma_4\gamma_1\gamma_2$ The special 5-cycle at c is given by $\gamma 4\gamma_1\gamma_2\gamma_3$

3)



The special 1-cycle at a is given by $\alpha_1\alpha_2\alpha_3$

The special 2-cycle at a is given by $\alpha_2\alpha_3\alpha_1$

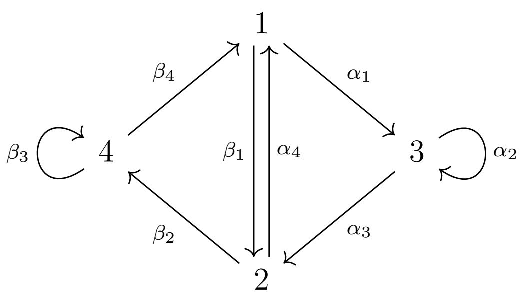
The special 3-cycle at a is given by $\alpha_3\alpha_1\alpha_2$

The special 1-cycle at b is given by $\beta_1\beta_2\beta_3$

The special 2-cycle at b is given by $\beta_2\beta_3\beta_1$

The special 3-cycle at b is given by $\beta_3\beta_1\beta_2$

4)



This is similar to 1), but there is a loop on both sides now. The special 1-cycle at a is given by $\alpha_1\alpha_2\alpha_3\alpha_4$.

The special 3-cycles at a are given by $\alpha_2\alpha_3\alpha_4\alpha_1$ and $\alpha_3\alpha_4\alpha_1\alpha_3$.

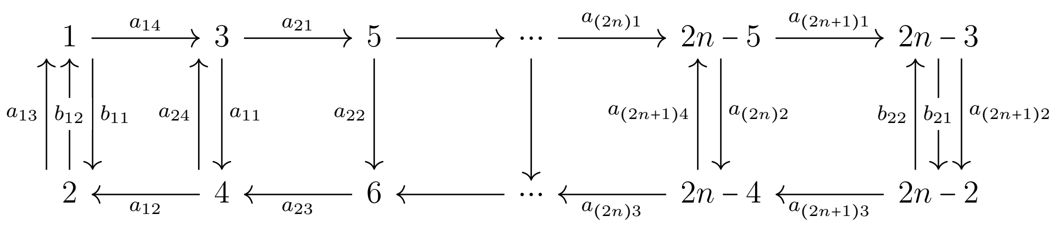
The special 2-cycle at a is given by $\alpha_4\alpha_1\alpha_2\alpha_3$.

The special 1-cycle at b is given by $\beta_1\beta_2\beta_3\beta_4$

The special 4-cycles at b are given by $\beta_3\beta_4\beta_1\beta_2$ and $\beta_4\beta_1\beta_2\beta_3$.

The special 2-cycle at b is given by $\beta_2\beta_3\beta_4\beta_1$.

5)



If we call each square formed in the quiver $\Box_i$, then the special cycles for each vertex $v \in B_0^{(5)}$ are the cycles $a_{i1}a_{i2}a_{i3}$ for each $i = 1, 2, \cdots, 2n+1$, and

$b_{12}b_{11}$ and $b_{22}b_{21}$

3.0.3 Set of relations and definition of Brauer graph algebras

Here, we define what is called an ideal of relations I_B in KQ_B, the path algebra of Q_B. This ideal is generated by three types of relations. For this we recall that we identify the set of edges E of a Brauer graph B with the set of vertices Q_0 of the corresponding quiver Q_B and that we denote the set of vertices of the Brauer Graph by B_0. Also, when we say relation, we mean

Relations of type I:

$$C_v^{m(v)} - C_{v'}^{m(v')},$$

for any $i \in Q_0$ and for any two distinct special i-cycles $C_v^{m(v)}$ and $C_{v'}^{m(v')}$ such that both v and v' are not truncated. In other words, $m(v)\mathrm{val}(v) \neq 1$ and $m(v')\mathrm{val}(v') \neq 1$.

Relations of type II:

$$C_v^{m(v)}\alpha_1$$

for any $i \in Q_0$ and any $v \in V$ and where $C_v^{m(v)}$ is any special i-cycle $C_v = \alpha_1\alpha_2...\alpha_n$.

Relations of type III:

$$\alpha\beta$$

for any $\alpha, \beta \in Q_1$ such that $\alpha\beta$ is not a subpath of any special cycle except if $\alpha = \beta$ is a loop associated to a vertex v of valency one and multiplicity $m(v) > 1$.

The algebra $A_B = KQ_B/I_B$ is called the *Brauer graph algebra* associated to the Brauer graph G

Example 3.0.6. Sets of relations for the examples in 2.2

1. Set of the three types of relations defining the Brauer graph algebra $A_1 = KQ_{B^{(1)}}/I_{B^{(1)}}$:

   Type I: $\alpha_0\alpha_1\alpha_2\alpha_3 - \alpha_1\alpha_2\alpha_3\alpha_0, \alpha_2\alpha_3\alpha_0\alpha_1 - \beta_1\beta_2\beta_3, \alpha_1\alpha_2\alpha_3\alpha_0 - \beta_2\beta_3\beta_1$

Type II: $\alpha_0\alpha_1\alpha_2\alpha_3\alpha_0, \alpha_1\alpha_2\alpha_3\alpha_0\alpha_1, \alpha_2\alpha_3\alpha_0\alpha_1\alpha_2, \beta_1\beta_2\beta_3\beta_1, \beta_2\beta_3\beta_1\beta_2, \beta_3\beta_1\beta_2\beta_3$

Type 3: $\alpha_2\beta_3, \alpha_1\beta_1, \alpha_1\beta_3, \beta_2\alpha_3, \beta_3\alpha_3$

2. Set of the three types of relations defining the Brauer graph algebra $A_2 = KQ_{B^{(2)}}/I_{B^{(2)}}$:

   Like in [11], we follow the notation of writing $\alpha_i\alpha_{i+1}\alpha_{i+2}\alpha_{i+3}$ for example as α^4.

   Type I: $\alpha_1\alpha_2\alpha_3 - \beta_1\beta_2\beta_3$, $\alpha_2\alpha_3\alpha_1 - \beta_2\beta_3\beta_1$,$\alpha_3\alpha_1\alpha_2 - \beta_3\beta_1\beta_2$

   Type II: α^5, β^5

   Type III: $\alpha_1\gamma_1, \gamma_1\beta_3, \beta_3\alpha_1, \alpha_3\beta_1, \beta_1\gamma_3, \gamma_3\alpha_3, \alpha_2\gamma_4, \gamma_4\alpha_2$

3. Set of the three types of relations defining the Brauer graph algebra $A_3 = KQ_{B^{(3)}}/I_{B^{(3)}}$:

   Type I: $\alpha_1\alpha_2\alpha_3 - \beta_1\beta_2\beta_3$, $\alpha_2\alpha_3\alpha_1 - \beta_2\beta_3\beta_1$, $\alpha_3\alpha_1\alpha_2 - \beta_3\beta_1\beta_2$

   Type II: α^4, β^4

   Type III: $\alpha_1\gamma_4, \gamma_4\alpha_3, \gamma_1\beta_3, \beta_3\alpha_1, \gamma_2\beta_2, \beta_2\gamma_2, \alpha_3\beta_1, \beta_1\gamma_3$

4. Set of the three types of relations defining the Brauer graph algebra $A_3 = KQ_{B^{(3)}}/I_{B^{(3)}}$:

   Type I: $\alpha_1\alpha_2\alpha_3\alpha_4 - \beta_1\beta_2\beta_3\beta_4$, $\alpha_2\alpha_3\alpha_4\alpha_1 - \alpha_3\alpha_4\alpha_1\alpha_2, \alpha_4\alpha_1\alpha_2\alpha_3 - \beta_2\beta_3\beta_4\beta_1, \beta_3\beta_4\beta_1\beta_2 - \beta_4\beta_1\beta_2\beta_3$.

   Type II: α^5, β^5

   Type III: $\beta_1\alpha_4, \alpha_4\beta_1, \alpha_3\beta_2, \beta_4\alpha_1$

5. Set of the three types of relations defining the Brauer graph algebra $A_3 = KQ_{B^{(4)}}/I_{B^{(4)}}$:

   Type I: $b_{11}b_{12} - b_{22}b_{21}, a_{i1}a_{i2}a_{i3} - a_{j1}a_{j2}a_{j3}$, for $i, j = 1, 2 \cdots 2n-2, j \neq i$,

   Type II: a^4, b^3

   Type III: $b_{12}a_{14}, b_{11}a_{13}, a_{13}b_{11}, b_{21}a_{(2n+1)3}, b_{22}a_{(2n+1)2}$

Brauer Graph Algebras form a very important class of algebras, and we show later on that the class of Brauer Graph Algebras and symmetric special biserial algebras coincide. We show these characterisations in a step-by-step manner. First we show that Brauer Graph Algebras are symmetric.

Theorem 3.0.7. *Brauer Graph Algebras are symmetric algebras*

Proof. First, we show that the ideal I_B of KQ_B is admissible. Every special cycle $C_v^{m(v)}$ is of the form $\alpha_1\alpha_2\alpha_3\cdots\alpha_k$, where $s(\alpha_1) = t(\alpha_k)$ and $k > 1$. Here the length of $C_v^{m(v)}$, denoted $l(C_v^{m(v)}) = k$. Therefore $I_B \subseteq \mathcal{R}^2$.

Now we show $\mathcal{R}^n \subseteq I_B$. Take $n = \max_{v \in Q_0} l(C_v^{m(v)})$, and take any path p with $l(p) \geq n+1$. Clearly, for the relations of type III, the condition is fulfilled, since the length of every path in the relation in type III is 2. For the relation of type II, we have that $l(C_v^{m(v)}\alpha_1) = k+1 \leq l(p)$. So, $p \in I_B$. For the type III relations however, we know that every path p with $l(p) \geq n+1$ is of the form $\beta_1\beta_2\cdots\beta_t$, with $t \geq n+1$. p can be rewritten as

$$\beta_1\beta_2\cdots\beta_t = (\alpha_i\cdots\alpha_k - \alpha_j\cdots\alpha_k)\beta_1\beta_2\cdots\beta_{t'} = (C_v^{m(v)} - C_{v'}^{m(v')})p'$$

where $s((C_v^{m(v)}) = t((C_v^{m(v)}), s(C_{v'}^{m(v')}) = t(C_{v'}^{m(v')})$ and $p' = \beta_1\beta_2\cdots\beta_{t'}$ such that $l(p') = l(p) - (l(C_v^{m(v)}) - l(C_{v'}^{m(v')}))$. So $p \in I_B$, and hence we get that $\mathcal{R}^n \subseteq I_B$ for $n = \max_{v \in Q_0} l(C_v^{m(v)})$

Now we show that the algebra is symmetric, that is we show that there exists a symmetric linear form $f : A_B \to K$ such that $\mathrm{Ker}(f)$ does not contain any non-zero left ideal of A_B. Define the map f as follows:

$$f(p) = \begin{cases} 1 & \text{, if } p = C_v^{m(v)} \text{ for some } v \in B_0 \\ 0 & \text{, otherwise} \end{cases}$$

Then the only way there is a zero ideal of A_B in Ker$f$ is if $f(p) = 0$, which would mean that there are no special cycles in the quiver Q_B, which is impossible. Hence, A_B is symmetric □

Remark 3.0.8. A Brauer Graph Algebra is indecomposable if and only if its corresponding Brauer Graph is connected.

Theorem 3.0.9. *Brauer Graph Algebras are special biserial*

Proof. We need to check that the condtions (G1) and (G2) for the BGA KQ_B/I_B are fulfilled.

(G1) is true since every edge is connected to exactly two vertices in B, and therefore, the cyclic ordering around those two vertices, will give either exactly two arrows going in, or exactly two vertices going out of the vertex i in the quiver Q_B.
For the condition (G2), if B is a Brauer graph with only two edges, then the quiver will consist of only two vertices, and the result follows. So, let B be a Brauer Graph with $|B_1| > 2$. Then, from the relation of type III, we have that there will always be a path $\alpha\beta$ in I_B which is not a subpath of any special cycle. So for every arrow α_1, there will always be a continuation $\alpha_1\alpha_2$ such that $\alpha_1\alpha_2$ is not part of any special cycle by definition of a special cycle with the number of edges more than 2 in the Brauer Graph. □

Corollary 3.0.10. *BGAs are of tame representation type*

Proof. Since BGAs are special biserial, and special biserial algebras are tame, therefore BGAs are tame. □

Theorem 3.0.11. *[13] A bound quiver algebra is a symmetric special biserial algebra if and only if it is a Brauer Graph Algebra*

3.0.4 Indecomposable Projective modules over Brauer Graph Algebras

In this section we compute the indecomposable projective modules over the Brauer Graph Algebras presented in Examples 3.0.5.

First we state some important properties of the indecomposable projective modules over a symmetric special biserial algebra, and we begin with a lemma.

Lemma 3.0.12. *Every indecomposable projective module over a path algebra of type $\mathbb{A}$ with linear orientation is uniserial*

Proposition 3.0.13. *The projective indecomposable modules over a symmetric special biserial algebra are either uniserial or biserial.*

Proof. Let Λ be a special biserial algebra, and let $P_i = e_i\Lambda$ be the indecomposable projective module at some vertex i of the quiver.

First we recall that the (G2) condition for special biserial algebras is that for every arrow α, there exists at most one β such that $\alpha\beta \notin I$ and at most one arrow γ such that $\gamma\beta \notin I$. This means that there is at most one path entering and leaving α that can live in the quotient algebra KQ/I. So for every indecomposable projective module P_i with vertex i, this means that starting from i, there can be either one or two paths starting from i.
Call these two paths $p = i \xrightarrow{\alpha_1} i_1 \xrightarrow{\alpha_2} \dots \xrightarrow{\alpha_t} i_t$ and $p' = i \xrightarrow{\alpha'_1} j_1 \xrightarrow{\alpha'_2} \dots \xrightarrow{\alpha'_s} j_s$ with $s(p) = s(p') = i$, and let us take the arrow α with $t(\alpha) = i$. Let us denote the modules generated by the paths p and p', U_1 and U_2 respectively. If none of these paths is not in the ideal I, then rad $P_i = U_1 + U_2$, where U_1 and U_2 are both uniserial, by lemma 3.0.12. If either $\alpha\alpha_1 \notin I$ or $\alpha\alpha'_1 \notin I$, then rad $P_i = U_1$ or U_2, thus making P_i either uniserial or biserial. □

Recall that the **descending Loewy series** of an indecomposable projective module over a $K-$ algebra is the following composition series

$$M = \operatorname{rad}^0 M \supseteq \operatorname{rad}^1 M \supseteq \operatorname{rad}^2 M \supseteq \cdots \supseteq \operatorname{rad}^n M = 0.$$

where each quotient $\operatorname{rad}^i M/\operatorname{rad}^{i+1} M$ is simple.

Example 3.0.14. Let A_1, A_2, A_3, A_4 be the Brauer Graph Algebras presented in Examples 3.0.5 given by the Brauer Graphs $B^{(1)}, B^{(2)}, B^{(3)}, B^{(4)}$. Then the projective indecomposable modules are:

1. $$P_1 :: \begin{array}{ccc} & 1 & \\ 1 & & 2 \\ 2 & & 3 \\ 3 & & 1 \\ & 1 & \end{array} \qquad P_2 : \begin{array}{ccc} & 2 & \\ 3 & & 4 \\ 1 & & 3 \\ 1 & & \\ & 2 & \end{array} \qquad P_3 : \begin{array}{ccc} & 3 & \\ 1 & & 2 \\ 1 & & 4 \\ 2 & & \\ & 3 & \end{array} \qquad P_4 : \begin{array}{c} 4 \\ 3 \\ 2 \\ 4 \end{array}$$

   Here rad P_1 is the representation $\begin{array}{ccc} 2 & & 4 \\ 5 & & 3 \\ & 1 & \end{array}$, $\operatorname{rad}^2 P_1$ is the representation $\begin{array}{ccc} 5 & & 3 \\ & 1 & \end{array}$ and so on. The Loewy series for P_1 is $P_1 = \operatorname{rad}^0(P_1) \supseteq \operatorname{rad}^1(P_1) \supseteq \operatorname{rad}^2(P_1) \supseteq \operatorname{rad}^3(P_1) \supseteq \operatorname{rad}^4(P_1) = 0$

   For P_1, $\operatorname{rad} P_1/\operatorname{rad}^2 P_1$ is the semisimple module $S(1) \oplus S(2)$, $\operatorname{rad}^2 P_1/\operatorname{rad}^3 P_1$ is $S(2) \oplus S(3)$ and so on. As we can see, P_1 here is a biserial module as the radical "branches" out into two chains of submodules. P_4, however is uniserial as the radical does not branch out into two or more chains of submodules. That is, it's composition series is unique.

2. $$P_1 :: \begin{array}{ccc} & 1 & \\ 2 & & 4 \\ 5 & & 3 \\ & 1 & \end{array} \qquad P_2 : \begin{array}{ccc} & 2 & \\ 5 & & 3 \\ 1 & & 4 \\ & & 5 \\ & 2 & \end{array} \qquad P_3 : \begin{array}{ccc} & 3 & \\ 1 & & 4 \\ 4 & & 5 \\ & & 2 \\ & 3 & \end{array} \qquad P_4 : \begin{array}{ccc} & 4 & \\ 5 & & 3 \\ 2 & & 1 \\ 3 & & \\ & 4 & \end{array}$$

3. $$P_1 :: \begin{array}{ccc} & 1 & \\ 2 & & 4 \\ 5 & & 3 \\ & 1 & \end{array} \qquad P_2 : \begin{array}{ccc} & 2 & \\ 5 & & 5 \\ 1 & & 3 \\ & & 4 \\ & 2 & \end{array} \qquad P_3 : \begin{array}{ccc} & 3 & \\ 4 & & 1 \\ 2 & & 4 \\ 5 & & \\ & 1 & \end{array} \qquad P_4 : \begin{array}{ccc} & 4 & \\ 2 & & 3 \\ 5 & & 1 \\ 3 & & \\ & 4 & \end{array}$$

4. $P_1 :: \begin{array}{ccc} & 1 & \\ 3 & & 2 \\ 3 & & 4 \\ 2 & & 4 \\ & 1 & \end{array}$ $P_2 : \begin{array}{ccc} & 2 & \\ 4 & & 1 \\ 4 & & 3 \\ 1 & & 3 \\ & 2 & \end{array}$ $P_3 : \begin{array}{c} 3 \\ 3 \\ 2 \\ 4 \\ 3 \end{array}$ $P_4 : \begin{array}{c} 4 \\ 4 \\ 1 \\ 2 \\ 4 \end{array}$

5. Here, the indecomposable projective modules P_i at i are uniserial for $i = 3, 4, \cdots 2n-4$, and for $i = 1, 2, 2n-3, 2n-2$, they are biserial.

Remark 3.0.15. We have defined ribbon graphs and ribbon data in Chapter 2. In fact, Brauer Graphs can be considered as ribbon graphs where the multiplicity of the vertices of the Brauer Graphs can be given by the weights of the vertices of the Ribbon Graph

3.0.5 Gentle Algebras and Brauer Graphs: The connection

Gentle Algebras and Brauer Graph Algebras are two very important classes of algebras that have been studied for a long time. They are both contained inside the class of special biserial algebras. Here, we show that there is a relation between Gentle Algebras and Brauer Graph Algebras: The trivial extension and admissible cuts.

We have that in 2.3.3 that the trivial extension of a finite dimensional algebra is symmetric. Combined with the fact that A is gentle if and only if $T(A)$ is special biserial(2.3.5), we get that $T(A)$ is a Brauer Graph Algebra. We state this in the form of a theorem and we explore and document this fact with the help of a few examples.

Theorem 3.0.16. *If A is a gentle algebra, then its trivial extension $T(A)$ is a Brauer Graph Algebra*

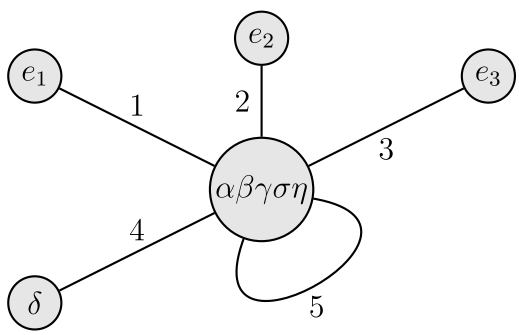
Example 3.0.17. 1. Take the following bound quiver (Q, I):

$$\begin{tikzcd}
 & & 3 \arrow[dr, "\gamma"] & & \\
1 \arrow[r, "\alpha"] & 2 \arrow[ur, "\beta"] & & 5 \arrow[dl, "\eta"] \arrow[loop right, "\sigma"] \\
 & & 4 \arrow[ul, "\delta"] & &
\end{tikzcd}$$

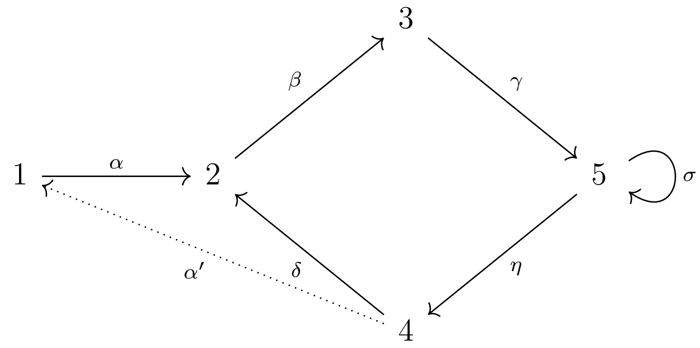
where $I =< \delta\beta, \gamma\eta, \sigma^2 >$. Here, I is admissible by taking $n = 5$ for the arrow ideal $\mathcal{R}^n$.

Also, the quiver is gentle from the fact that the quiver is gentle at each vertex $i \in Q_0$. Here, by being gentle at each vertex we mean that i as well as the arrows adjacent to the vertex i obey the conditions (G1),(G2),(G3),(G4), at a maximal configuration. Let $A = KQ/I$. Then the trivial extension of A is $T(A) = A \rtimes D(A)$, where $D(A) = D(KQ/I) = \mathrm{Hom}_K(KQ/I, K)$,

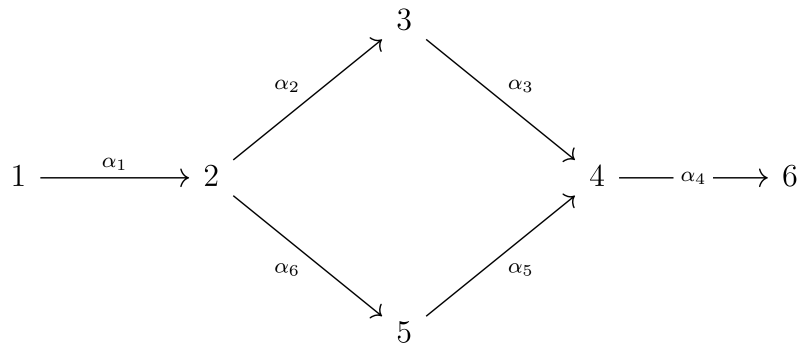
We have to show that $T(A)$ is Brauer graph algebra. For this we need to construct the graph Γ_A of the given gentle algebra. Here, the extended maximal path set $\overline{\mathcal{M}} = \{p_1 = \alpha\beta\gamma\sigma\eta\delta, e_1, e_2, e_3\}$. There is an edge E_i between the vertices m and n_i of the graph if two maximal paths m and n_i pass through the same vertex i, where i is a vertex in the maximal path m.So here, we see that there is an edge between the vertices. e_1 and $\alpha\beta\gamma\sigma\eta$. This algorithm is already mentioned in chapter 2, so following that, we get the following graph



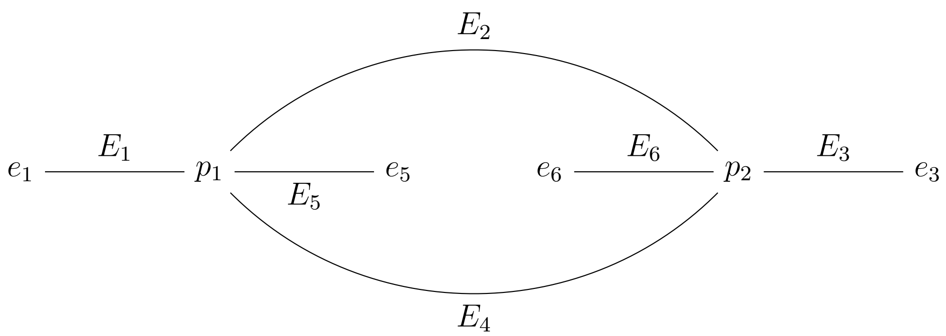
The extended quiver of the above quiver is the quiver $\overline{Q} = (Q_0, \overline{Q_1}\ )$ with the set of vertices being Q_0 itself and the set of arrows $\overline{Q}_1 = Q_1 \cup Q_1'$, where $Q' = \{\beta_m|$ for every $m \in M$ where $s(_m) = t(m)$ and $t(_m) = s(m)\}$ which is the following quiver with the extra arrow α' indicated by dotted lines.



2. Another example is the following gentle algebra as in Fig 1



with the following ribbon graph



and the following extended quiver $\overline{Q}$

$$\begin{tikzcd}
& & 3 \arrow[dr, "\alpha_3"] & & \\
1 \arrow[r, "\alpha_1"] & 2 \arrow[ur, "\alpha_2"] \arrow[dr, "\alpha_5"'] & & 4 \arrow[r, "\alpha_4"] \arrow[lll, dotted, bend left=60, "\alpha_7"] & 6 \arrow[lll, dotted, bend right=60, "\alpha_8"'] \\
& & 5 \arrow[ur, "\alpha_6"'] & &
\end{tikzcd}$$

Theorem 3.0.18. *If $A = KQ/I$ is a gentle algebra with the associated ribbon graph being Γ_A, then $T(A)$ is the Brauer Graph Algebra with Brauer Graph being Γ_A with multplicities of the vertices all being 1.*

Remark 3.0.19. $\overline{Q}$ is the quiver of the Brauer graph algebra with associated Brauer graph Γ_a and with cyclic ordering of the edges of Γ_A around each vertex induced by the arrows in $\overline{Q}$

Examples 3.0.17 indeed shows that the above remark is true if we follow the quiver construction from a Brauer Graph as in section 3.0.2

Corollary 3.0.20. *If A is a gentle algebra arising from a triangulation T of an oriented marked surface with marked points in the boundary as defined in chapter 2, then $\Gamma_A = T$ as ribbon graphs*

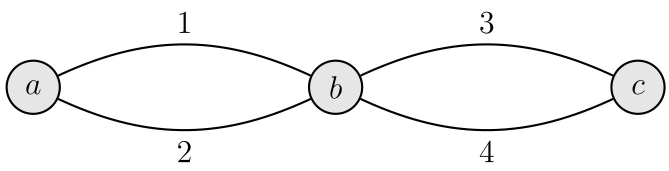
3.0.6 Slices

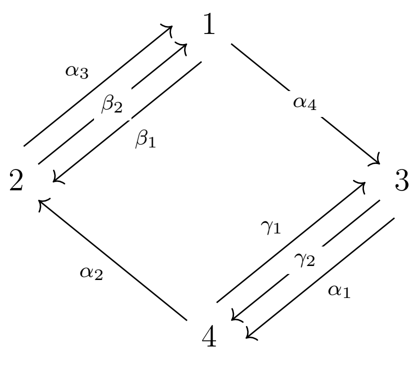
In the previous section we obtained a new "operation" to go from a gentle algebra to a Brauer Graph Algebra via the trivial extension. Now we look at the converse of that idea: Slices(or admissible cuts as in [13, 11]). While the trivial extension of a gentle algebra is a way of obtaining a new algebra by increasing the dimension of the underlying vector space, *slices* do the opposite-we reduce the dimension of the vector space to obtain another algebra, that is we go from a Brauer Graph Algebra to a gentle algebra. This section is where we study the notion of admissible cuts using specific examples.

Definition 3.0.21. Let $A = KQ/I$ be a Brauer graph algebra with Brauer graph G and multiplicity function m identically equal to one. A slice Σ of Q is a set of arrows containing exactly one arrow from every special cycle C_v from its respective permutation class, where $\mathrm{val}(v) > 1$, for some $v \in B_0$

The *sliced* algebra with the slice Σ is the algebra $A_\Sigma = KQ_\Sigma/\langle I \cup \Sigma\rangle$ where $\langle I \cup \Sigma\rangle$ is the ideal generated by $I \cup \Sigma$ in KQ_Σ

Example 3.0.22. 1. Let us take the Brauer Graph Algebra with the following Brauer graph and corresponding quiver





Let $\Sigma_1 = \{\alpha_1, \beta_1, \gamma_1\}$, and let $\Sigma_2 = \{\alpha_2, \gamma_2, \beta_2\}$. Then we have the sliced algebra with the slice Σ_1 to be the algebra $A_{\Sigma_1} \cong K(Q/\Sigma_1)/\langle \beta_2\alpha_4, \alpha_4\gamma_2, \gamma_2\alpha_2, \alpha_2\beta_2\rangle$ and the sliced algebra $A_{\Sigma_2} \cong K(Q/\Sigma_2)/\langle \alpha_3\beta_1, \alpha_1\gamma_1\rangle$

Chapter 4

Future Scope: Derived Equivalence and Geometric Models of Gentle Algebras

In this thesis, we looked at algebraic objects called algebras, more specifically special types of algebras called path algebras, and we have studied the modules over these path algebras, also called as representations.

Representations of a quiver Q are a very combinatorial way of understanding the representation theory of the finite dimensional path algebra KQ/I, as studying them is the same as studying the modules over the same path algebra. This is because there is an equivalence

$$\text{Rep } Q \simeq \text{Mod } KQ$$

which restricts to an equivlance rep $Q \simeq$ mod KQ as well as an equivalence

$$\text{Rep } (Q.I) \simeq \text{Mod } KQ/I$$

which restricts to an equivalence rep $(Q, I) \simeq$ mod KQ/I

Gentle algebras are a well-understood class of algebras, the study of which is very accessible. As studied as they already are, they are still a very good area for testing of new ideas, and still have a lot of potential for a great amount of future research, like studying the derived category of these gentle algebras, and consequently the geometric models of the module category of gentle algebras, as well as the geometric models of the derived category of gentle algebras.

4.1 Derived Equivalence of Gentle Algebras

As it turns out(unsurprisingly), category theory, and consequently homological algebra, has a lot of importance in representation theory. More specifically, if $\mathcal{D}(A)$ denotes the bounded derived category of finitely generated $A-$ modules, where A is a gentle algebra then the geometric models attached to the *module category* as well as the *bounded derived category* of finitely generated A-modules have a boatload of potential research possibilities. So, essentially in this section we will want to build a category which satisfies certain properties.

Now let $\mathcal{D}^b(\mathscr{A})$ denote the bounded derived category of $\mathscr{A}$. Then we have the following theorem

Theorem 4.1.1. *[11, 6]Suppose (S, M) is a marked oriented surface with the marked points lying on the boundary of S. Let A_T and $A_{T'}$ be the gentle algebras that arise from any choice of triangulations T and T' respectively, and let $\mathscr{A}_T$ and $\mathscr{A}_{T'}$ be the categories of finitely generated A_T and $A_{T'}$ modules respectively, then*

$$\mathcal{D}^b(\mathscr{A}_T) \simeq \mathcal{D}^b(\mathscr{A}_{T'})$$

And the result from Rickard[8] highlights one more relation between these algebras.

Theorem 4.1.2. *Let T and T' be triangulations of (S, M) having the same parameters, and let A_T and $A_{T'}$ denote the corresponding gentle algebras. If*

$$\Lambda_T = A_T \ltimes D(A_T) \quad \textit{and} \quad \Lambda_{T'} = A_{T'} \ltimes D(A_{T'}),$$

are the associated Brauer graph algebras, then

$$\mathcal{D}^b(\mathscr{B}_T) \cong \mathcal{D}^b(\mathscr{B}_{T'})$$

, *where $\mathscr{B}$ denotes the category of finitely generated* $\Lambda-$*modules*

The proof for these theorems are omitted for now, and instead we work with examples to verify said theorems. So, let us take the following examples:

Example 4.1.3. 1. Let us take the algebras A_T and $A_{T'}$ obtained from the triangulations of the following octagon

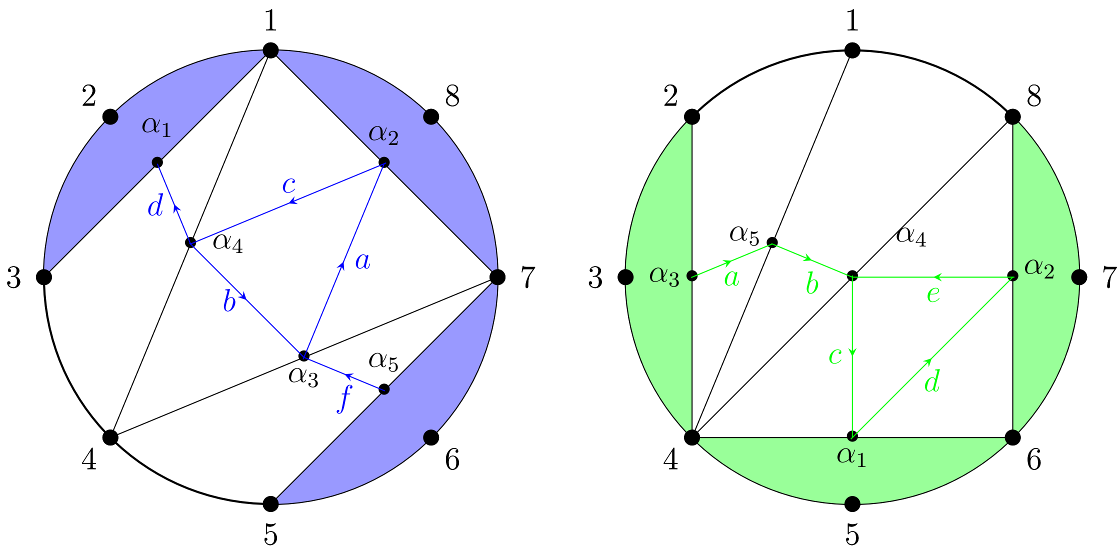
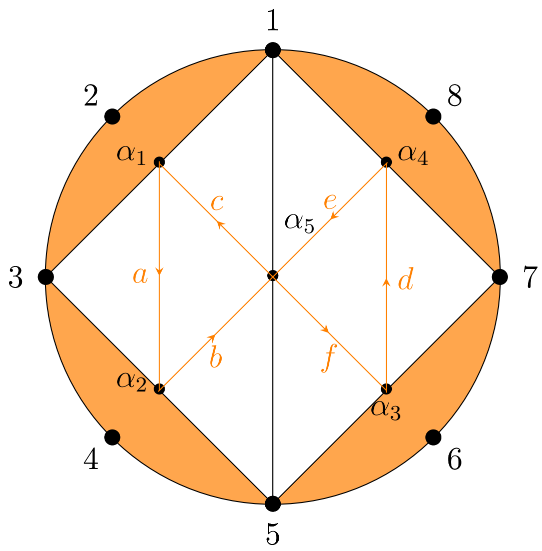


Figure 1: Triangulations T and T' and the quivers Q_T and $Q_{T'}$

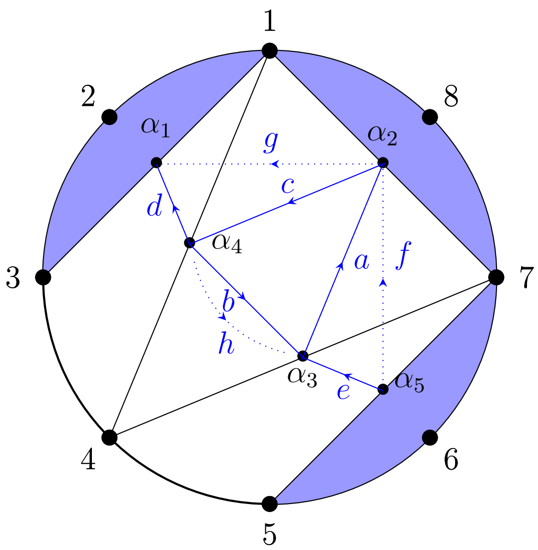
The two triangulations of the octagon in the above figure have the same parameters, as one can check. In fact, by remark 3.21 of [11], since they have the same number of boundary triangles incident on the single boundary component, we get that the two algebras obtained are derived equivalent,i.e. $\mathcal{D}^b(\mathscr{A}_T) \simeq \mathcal{D}^b(\mathscr{A}_{T'})$. The necessary boundary triangles are highlighted in blue in T' and in green in T''.

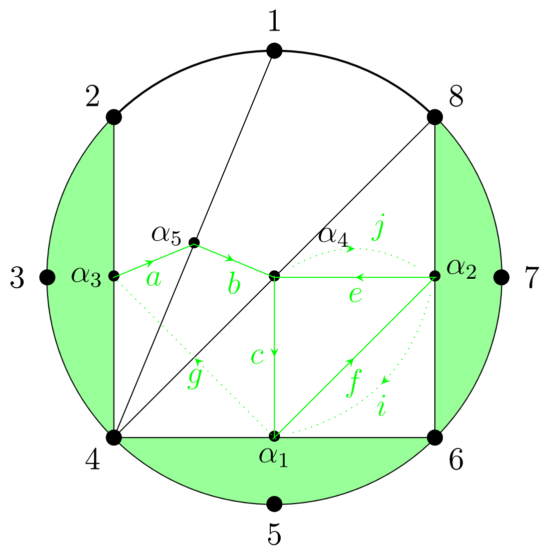
However the following algebra obtained is not derived equivalent to any of the above algebras, since it has 4 boundary triangles, which are highlighted in orange in the following triangulation T''



Therefore we have that $\mathcal{D}^b(\mathscr{A}_T) \simeq \mathcal{D}^b(\mathscr{A}_{T'}) \not\simeq \mathcal{D}^b(\mathscr{A}_{T''})$

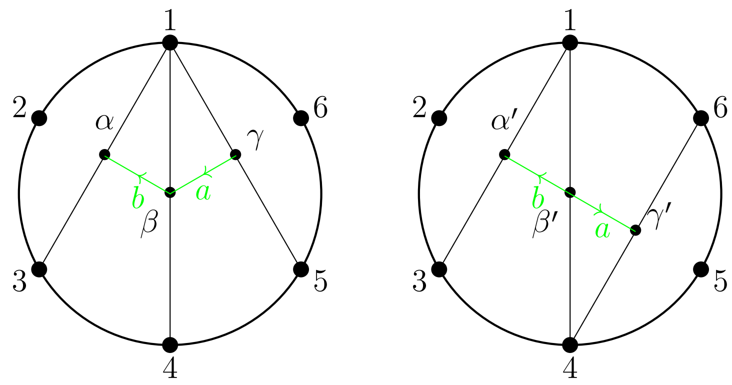
Now we take the trivial extensions of the algebras A_T and $A_{T'}$,, and we have the following extended quivers



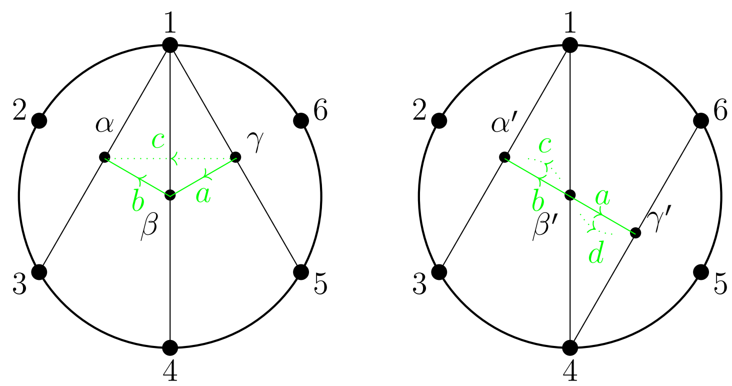


Clearly, their trivial extensions are also bounded derived equivalent since the number of boundary triangles is unchanged.

2. The following are triangulations of hexagons, the algebras of which are derived equivalent since the number of boundary triangles is the same



The trivial extensions with the extended quivers are as follows



4.2 Geometric Models of the Module Category of Gentle Algebras

Let $A = KQ/I$ be a gentle algebra. The indecomposable finite-dimensional A-modules are described in terms of string modules and band modules. Thus, the module category $\operatorname{mod} A$ is governed by the string and band combinatorics of the bound quiver (Q, I).

A geometric interpretation of this combinatorics is given by the model of Baur and Coelho Simões. In their work, gentle algebras are realised as tiling algebras associated to partial triangulations of unpunctured surfaces with marked points on the boundary. This gives a surface model for the module category of a gentle algebra.

Theorem 4.2.1 (Baur–Coelho Simões [3, Theorems 3.8, 3.9 and 3.15])**.** *Let A_P be* *the tiling algebra associated to a partial triangulation P of an unpunctured marked surface (S, M). Then the following correspondences hold:*

1. *equivalence classes of non-trivial permissible arcs in (S, M) correspond to nonzero strings of A_P;*

2. *permissible closed curves in (S, M) correspond to powers of bands of A_P;*

3. *certain elementary moves of permissible arcs correspond to irreducible morphisms between the associated string modules.*

This theorem suggests a concrete direction for further study. For the gentle algebras appearing in this thesis, one may construct the corresponding marked surface and partial triangulation, and then compare the algebraic data of (Q, I) with the geometric data on the surface. In particular, a string w in the bound quiver gives a string module $M(w)$, while the geometric model associates to it a permissible arc. Conversely, a permissible arc determinesa string by recording its intersections with the partial triangulation.

Thus, the classification of string modules can be translated into the classification of permissible arcs. Similarly, band modules are related to permissible closed curves. Moreover, irreducible morphisms between string modulescan be studied through elementary moves of the corresponding arcs. This gives a geometric method for studying parts of the Auslander–Reiten theory of gentle algebras.

Problem 4.2.2. For each gentle algebra $A = KQ/I$ constructed in this thesis, construct the associated marked surface and partial triangulation. Then describe explicitly how strings, bands, and irreducible morphisms are represented by permissible arcs, permissible closed curves, and elementary moves of arcs.

4.3 Geometric Models of the Derived Category of Gentle Algebras

The module category $\operatorname{mod} A$ is only one level of the representation theory of a gentle algebra A. A finer invariant is the bounded derived category

$$\mathcal{D}^b(A) = \mathcal{D}^b(\operatorname{mod} A),$$

whose objects are bounded complexes of finite-dimensional A-modules. For gentle algebras, the indecomposable objects in this category are described algebraically by homotopy strings and homotopy bands.

Opper, Plamondon, and Schroll give a geometric model for the derived category of a gentle algebra. Starting from the ribbon graph associated to a gentle algebra, one constructs a marked surface S_A. In this surface, suitable curves encode the indecomposable objects of the derived category.

Theorem 4.3.1 (Opper–Plamondon–Schroll [7, Theorem 2.13, Theorem 3.3, Theorem 4.1, and Corollaries 5.2, 5.4])**.** *Let A be a finite-dimensional gentle algebra, viewed with the trivial grading, and let S_A be the marked surface associated to its ribbon graph. Then indecomposable objects of $\mathcal{D}^b(A)$ are represented by curves on S_A. Moreover, intersections of curves give basis morphisms, resolutions of crossings describe mapping cones, and the Auslander–Reiten translate of a perfect object is realised by rotating endpoints along the boundary.*

This result gives a geometric way to study derived categories of gentle algebras. Homotopy strings and homotopy bands may be studied through curves and closed curves, while morphisms and mapping cones may be studied through intersections and resolutions of crossings. Therefore, for the gentle algebras considered in this thesis, a natural next step is to construct the surface S_A explicitly and identify the curves corresponding to selected complexes.

Problem 4.3.2. For a gentle algebra A arising from the examples in this thesis, construct the associated marked surface S_A. Then describe explicitly how homotopy strings, homotopy bands, morphisms, and mapping cones are represented geometrically by curves, closed curves, intersections, and resolutions of crossings.

This direction can also be combined with derived-equivalence results for gentle algebras arising from triangulations. If two triangulations T and T' give derived equivalent gentle algebras A_T and $A_{T'}$, then one may ask how the corresponding geometric models are related under the equivalence

$$\mathcal{D}^b(A_T) \simeq \mathcal{D}^b(A_{T'}).$$

Appendix

5.1 Homotopy Category

Definition 5.1.1. Let R be a ring. A chain(cochain) complex C is a collection $\{C_n\}$(similarly $\{C^n\}$) of R–modules together with maps $d_n : C_n \to C_{n-1}$(respectively $d^n : C^n \to C^{n+1}$) such that the composition $d_{n-1} \circ d_n : C_n \to C_{n-2} = 0$, and respectively for cochain complexes, we have $d^{n+1} : C^{n+2} \to C^n$ is zero.

So a chain complex is of the following form

$$\cdots \to C_n \xrightarrow{d_n} C_{n-1} \xrightarrow{d_{n-1}} C_{n-2} \cdots$$

The above definition means that, for chain complexes, the image of d_n is contained in the kernel of d_{n-1}, and the notion is analogous for cochain complexes, and so we can talk about the quotient Ker d_{n-1}/Im d_n. This module is called the *n-th homology module* of the chain complex C, or the homology module at n, and we denote that module by $H_n(C)$. The maps d_i are called the differentials of C. Also, in this study we will look mainly at chain complexes, and we mention when we are working with cochain complexes otherwise.

Now that we have defined chain complexes, a natural question to ask is what if we have two chain complexes C and D, and we want to talk about the morphisms between them. This is what we define in the following

Definition 5.1.2. Let C and D be two chain complexes, and let $c_i : C_i \to C_{i-1}$ and $d_i : D_i \to D_{i-1}$ be the differentials of C and D respectively. A morphism, also called a chain map, of two chain complexes C and D is a sequence of maps $\{f_n : C_n \to D_n\}_{n \in \mathbb{Z}}$ such that $f_{n-1} \circ d_n = c_n \circ f_n$

So essentially it means the "squares" in the following diagram commutes.(When we say a diagram commutes it means that if we take a point in the diagram then we can arrive at any other point from any map or composition of maps, if possible)

Now that we have chain complexes, and we have morphisms between chain complexes, we have the category $\mathbf{Ch}(R)$, where the objects are chain complexes and the morphisms are chain maps between the two complexes.

The notion of homotopy comes from topology, and it is a way of relating two functions two topological spaces. We do a similar thing here, that is we will define a relation, called *chain homotopy*, which relates two chain maps between two different chain complexes, and thus *chain homotopy equivalence*. But first we begin by defining *null homotopy*, that is the homotopy is "null" in some sense.

Definition 5.1.3. A chain map f between two complexes C and D is called null homotopic if there exists maps $s_n : C_n \to D_{n-1}$ such that each f_n can be written as $f_n = d_n s_{n-1} + s_{n+1} c_n$.

Now we can look at the concept of *chain homotopy*

Definition 5.1.4. Two chain maps f and g are said to be chain homotopic if

$$f - g = ds + sd$$

That is, we have that $f - g$ is null homotopic.

Finally, we have chain homotopy equivalence

Definition 5.1.5. A chain map f between two chain complexes C and D is said to be a chain homotopy equivalence if there exists a chain map $g : D \to C$ such that gf and fg are chain homotopic to the identity map.

In other words, we have that $fg - \mathrm{id}_D = ds + sd$ and $gf - \mathrm{id}_C = ds + sd$

So upto now, we have seen the category $\mathbf{Ch}(R)$, the category where the objects were chain complexes of R-modules, and the morphisms were the chain maps. Now that we have defined chain homotopy equivalence, we now have a way of identifying two chain maps, and thus we can have a new category where the objects are the same as $\mathbf{Ch}(R)$, but then the morphisms will be the chain homotopy equivalence classes. So, if we denote the new category by $\mathcal{K}(R)$, then this category is the quotient category of $\mathbf{Ch}(R)$ by only the chain homotopy equivalence classes. That is, we leave the objects unchanged, and simply put, $\mathrm{Hom}_{\mathcal{K}(R)}(A,B)$ is $\mathrm{Hom}_{\mathbf{Ch}(R)}(A,B)/\sim$, where $\sim$ is the chain homotopy equivalence between two maps f and g.

Thus, we have our new category, which we will now call the *homotopy category*, where, again the objects are chain complexes of R-modules and the morphisms are chain homotopy equivalence class.

Remark 5.1.6. Note that the notion of two chain maps being chain homotopic, and chain homotopy equivalence are two different notions. That is, if two maps h and h' are chain homotopic, then they might not be chain homotopic equivalent. For example, let us take the zero chain map between two chain complexes. Clearly, 0 is chain homotopic to itself, but the zero map is not invertible, so clearly, it is not a chain homotopy equivalence. So the main takeaway is that two chain homotopic maps do not necessarily belong to the same chain homotopy equivalence class in $\mathcal{K}(R)$

5.2 Localisation of Categories

Localisation is the process of constructing a new category from a given category by formally inverting a specified class of morphisms [5, Chapter 1].

Let $\mathscr{C}$ be a category and let S be a subclass of the morphisms of $\mathscr{C}$. That is $S \subseteq \mathrm{Hom}_{\mathscr{C}}(A,B)$. The *localisation* of $\mathscr{C}$ with respect to S is a new category $\mathscr{C}[S^{-1}]$ together with a functor $Q : \mathscr{C} \to \mathscr{C}[S^{-1}]$ that satisfies the following criteria:

(L1) For every morphism σ in $\mathscr{C}$, we have that $Q\sigma$ is invertible in $\mathscr{C}[S^{-1}]$

(L2) If given another functor $F : \mathscr{C} \to \mathscr{D}$ such that $F\sigma$ is invertible, then we have a map $\tilde{F} : \mathscr{C}[S^{-1}] \to \mathscr{D}$, which means that $F = \tilde{F} \circ Q$, or in other words $\tilde{F}$ factors through Q

We can think of the image $Q(S)$ as the morphisms which are invertible. So here, $\mathscr{C}[S^{-1}]$ is a *universal object*, and localisation solves the universal localising problem, and the canocial functor is therfore unique upto isomorphism.

So in this section, we have defined what localisation means and have given two axioms for localisation, but we have not explicitly mentioned what the morphisms or objects are. So from here onward, we will describe the functor $Q : \mathscr{C} \to \mathscr{C}[S^{-1}]$, the objects of the category and the morphisms in more detail.

The objects of $\mathscr{C}[S^{-1}]$ are the same objects as that of $\mathscr{C}$. The *morphisms* of $\mathscr{C}[S^{-1}]$ are constructed as follows: We take the quiver Q with the vertices Q_0 being the objects of $\mathscr{C}$, and the class of arrows Q_1 being the disjoint union Mor $\mathscr{C} \bigsqcup S^{-}$, where the S^{-} is the class of formal inverses of the morphisms in $\mathscr{C}$. That is,

$$S^{-} = \{\sigma^{-} : Y \to X \mid S \ni \sigma : X \to Y\}.$$

We can think of Mor $\mathscr{C}$ being the usual arrow set Q_1, and S^{-} to be a subset of the arrow set of the opposite quiver Q^{op}.
Let $\mathcal{P}$ be the category of paths in this quiver, that is, a cateogry where the objects are paths and morphisms are the composition/concatenation of paths wherever possible and denote that by $\circ_{\mathcal{P}}$. Now after the construction of the objects, we can define the morphisms of $\mathscr{C}[S^{-1}]$ as the quotient of $\mathcal{P}$ modulo the following relations:

1. $\beta \circ_{\mathcal{P}} \alpha = \beta \circ \alpha$, where α, β are paths in the quiver, or we can also say that α and β are "composable" morphisms in Mor $\mathscr{C}$
2. $\mathrm{id}_{\mathcal{P}} X = \mathrm{id}_{\mathscr{C}} X$ for all $X \in$ Ob $\mathscr{C}$
3. $\sigma^{-} \circ_{\mathcal{P}} \sigma = \mathrm{id}_{\mathcal{P}} X$ and $\sigma \circ_{\mathcal{P}} \sigma^{-} = \mathrm{id}_{\mathcal{P}} Y$, for all $\sigma : X \to Y$ in S

So, in other words, the composition in $\mathcal{P}$ induces the composition in $\mathscr{C}[S^{-1}]$. The functor Q is the identity on objects but a morphism in $\mathscr{C}$ is sent to $\mathscr{C}[S^{-1}]$ via the following composition:

$$\text{Mor } \mathscr{C} \xrightarrow{\text{inc}} \text{Mor } \mathscr{C} \sqcup S^{-} \xrightarrow{\text{inc}} \mathcal{P} \xrightarrow{\text{can}} \text{Mor } \mathscr{C}[S^{-1}]$$

In this way, one can see the homotopy category $\mathcal{K}(R)$ to be the localised category of the category of chain complexes $\mathbf{Ch}(R)$, where the invertible morphisms is the set $Q(S)$, where S is the set of chain homotopy equivalence classes. Localisation is also a generalisation of the localisation of rings.

Definition 5.2.1. A morphism $C \to D$ of chain complexes is called a *quasi-isomorphism* if the maps $H_n(C) \to H_n(D)$ are all isomorphisms.

5.3 Derived Category

This section devoted to the derived category is purely from the perspective of a representation theorist, and as such we do not give a general definition of the derived category of an abelian category, since that requires the knowledge and prerequisites of derived functors, triangulated categories and so on.
Since, we are building up to what is called as *derived equivalence* of two finite-dimensional associative algebras, we would first have to know what triangulated categories are. So first, we define what the mapping cone of a morphism of two chain complexes C and D

Definition 5.3.1. Let $f : C \to D$. be a map of chain complexes. The mapping cone of f is the chain complex cone(f) whose degree n part is $C_{n-1} \oplus D_n$. In order to match other sign conventions, the differential in cone(f) is given by the formula

$$d_n^{\text{Cone}}(b, c) = (-d_{n-1}^C(b), d_n^D(c) - f_{n-1}(b)), (b \in C_{n-1}, , c \in D_n).$$

We check that $d_{n-1}^{\text{Cone}}(d_n^{\text{Cone}}(b, c)) = 0.$

Recall that

$$\text{Cone}(f)_n = C_{n-1} \oplus D_n,$$

with differential

$$d_n^{\text{Cone}}(b, c) = \left(-d_{n-1}^C(b),\, d_n^D(c) - f_{n-1}(b)\right),$$

where $b \in C_{n-1}$ and $c \in D_n$.

Let $(b,c) \in \mathrm{Cone}(f)_n$. Then

$$\begin{aligned} d^{\mathrm{Cone}}_{n-1} d^{\mathrm{Cone}}_n (b,c) &= d^{\mathrm{Cone}}_{n-1}\left(-d^C_{n-1}(b),\, d^D_n(c) - f_{n-1}(b)\right) \\ &= \left(-d^C_{n-2}\left(-d^C_{n-1}(b)\right),\, d^D_{n-1}\left(d^D_n(c) - f_{n-1}(b)\right) - f_{n-2}\left(-d^C_{n-1}(b)\right)\right) \\ &= \left(d^C_{n-2}d^C_{n-1}(b),\, d^D_{n-1}d^D_n(c) - d^D_{n-1}f_{n-1}(b) + f_{n-2}d^C_{n-1}(b)\right). \end{aligned}$$

Since C and D are chain complexes,

$$d^C_{n-2}d^C_{n-1} = 0, \qquad d^D_{n-1}d^D_n = 0.$$

Moreover, $f$is a chain map, so

$$d^D_{n-1} \circ f_{n-1} = f_{n-2} \circ d^C_{n-1}.$$

Therefore,

$$d^{\mathrm{Cone}}_{n-1} d^{\mathrm{Cone}}_n (b,c) = (0,0),$$

and hence

$$\left(d^{\mathrm{Cone}}\right)^2 = 0.$$

Let A be a gentle algebra and let $\mathscr{A} = A$−mod denote the category of finitely generated A−modules.

Definition 5.3.2. Let A be a gentle algebra and let $\mathscr{A} = A$−mod denote the category of finitely generated A−modules. Then the derived category of $\mathscr{A}$, denoted by $\mathcal{D}(\mathscr{A})$ is simply the localisation of the homotopy category $\mathcal{K}(A)$ at the class of quasi-isomorphisms. That is, $\mathcal{D}(A) = \mathscr{A}[I^{-1}]$, where I denotes the class of quasi-isomorphisms

Now we move on to the notion of bounded derived categories. First we say that a chain complex C is said to be *bounded below* if $C_n = 0$, for sufficiently small n and bounded above if $C_n = 0$ for sufficiently large n. C is said to be bounded if it is bounded above or bounded below.

Definition 5.3.3. The bounded derived category of $\mathcal{A}$, denoted by $\mathcal{D}^b(\mathscr{A})$ is the derived category $\mathcal{D}(\mathscr{A})$, where the chain complexes C in $\mathbf{Ch}(A)$ are bounded.

5.4 Derived Equivalence

While the category we were working with was an abelian category, that is, a nice category where finite products and coproducts existed, and the notion of kernels and cokernels exist for every morphism in the category, the derived category of an abelian category is not abelian anymore. It turns out to be a triangulated category. This structure is inherited from the fact that $\mathcal{K}(R)$ is triangulated. All of this theory is possible due to the work of Rickard[8, 9]

First, we define a tilting complex.

Definition 5.4.1. Let proj−A denote the category of finitely generated projective A−modules. Then a *tilting complex* is an object T of the bounded which satisfies the following conditions:

1. $\mathrm{Hom}(T, T[i]) = 0$ for all $i \neq 0$. Here, i is known as the translation functor(See [16])
2. $\mathrm{add}(T)$ generates $\mathcal{K}^b(\text{proj-}A)$ as a triangulated category. Here, $\mathrm{add}(T) = \{X \mid X \text{ is isomorphic to a direct summand of } T^{\oplus n} \text{ for some } n \geq 0\}$.(See, for example [1])

Now we define the notion of derived equivalence

Definition 5.4.2. Let T be a tilting complex of the bounded homotopy category $\mathcal{K}^b$(proj-A), and let A and B be two finite-dimensional associative algebras. Then A and B are said to be derived equivalent if $\mathcal{D}^b(\mathscr{A})$ and $\mathcal{D}^b(\mathscr{B})$ are equivalent as triangulated categories. Here, $\mathscr{A}$ and $\mathscr{B}$ denotes the category A−mod and B-mod respectively

By equivalent as triangulated categories, we mean that the triangulated structures between the two categories are preserved.

So in the derived category $\mathcal{D}^b(\mathscr{A})$, the triangles are precisely those triangles of the form

$$C \to D \to \mathrm{Cone}(f) \to C[1],$$

which play the same role in exact sequences as in abelian categories like $A-\mathrm{mod}$. Here [1] is the translation functor

[1] Ibrahim Assem, Daniel Simson, and Andrzej Skowroski. *Elements of the Representation Theory of Associative Algebras: Volume 1: Techniques of Representation Theory*. Elements of the Representation Theory of Associative Algebras. Cambridge University Press, 2006. ISBN: 9780521584234. URL: https://books.google.co.in/books?id=ayNHpi3tYhQC.

[2] Ibrahim Assem et al. *Gentle algebras arising from surface triangulations*. 2010. URL: https://projecteuclid.org/journals/algebra-and-number-theory/volume-4/issue-2/Gentle-algebras-arising-from-surface-triangulation....

[3] Karin Baur and Raquel Coelho Simões. “A geometric model for the module category of a gentle algebra”, *International Mathematics Research Notices* 2021.15 (2021), pp. 11357–11392.

[4] Ju. A. Drozd. “Tame and wild matrix problems”, *Representation Theory II*. Vol. 832. Lecture Notes in Mathematics. Berlin: Springer, 1980, pp. 242–258. DOI: 10.1007/BFb0088467.

[5] Henning Krause. *Homological Theory of Representations*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2021.

[6] Sefi Ladkani. *Derived Equivalence Classification of the Gentle Algebras Arising from Surface Triangulations*.

[7] Sebastian Opper, Pierre-Guy Plamondon, and Sibylle Schroll. *A geometric model for the derived category of gentle algebras*. 2025. arXiv: 1801.09659 [math.RT]. URL: https://arxiv.org/abs/1801.09659.

[8] Jeremy Rickard. “Derived categories and stable equivalence”, *Journal of Pure and Applied Algebra* 61.3 (1989), pp. 303–317. ISSN: 0022-4049. DOI: https://doi.org/10.1016/0022-4049(89)90081-9. URL: https://www.sciencedirect.com/science/article/pii/0022404989900819.

[9] Jeremy Rickard. “Morita Theory for Derived Categories”, *Journal of the London Mathematical Society* s2-39.3 (June 1989), pp. 436–456. ISSN: 0024-

6107. DOI: 10.1112/jlms/s2-39.3.436. eprint: https://academic.oup.com/jlms/article-pdf/s2-39/3/436/2470128/s2-39-3-436.pdf. URL: https://doi.org/10.1112/jlms/s2-39.3.436.

[10] Ralf Schiffler. *Quiver Representations*. CMS Books in Mathematics. Springer International Publishing, 2014. ISBN: 9783319092034. URL: https://books.google.co.in/books?id=0ctuBAAAQBAJ.

[11] Sibylle Schroll. *Brauer graph algebras*. 2018. DOI: 10.1007/978-3-319-74585-5_6. URL: https://doi.org/10.1007/978-3-319-74585-5_6.

[12] Sibylle Schroll. "Brauer graph algebras: a survey on Brauer graph algebras, associated gentle algebras and their connections to cluster theory", *Homological methods, representation theory, and cluster algebras*. Springer, 2018, pp. 177–223.

[13] Sibylle Schroll. *Trivial extensions of gentle algebras and Brauer graph algebras*. 2015. DOI: 10.1016/j.jalgebra.2015.07.037. URL: https://doi.org/10.1016/j.jalgebra.2015.07.037.

[14] Burkhard Wald and Josef Waschbüsch. "Tame biserial algebras", *Journal of Algebra* 95 (1985), pp. 480–500. URL: https://api.semanticscholar.org/CorpusID:122387589.

[15] Josef Waschbüsch and Andrzej Skowroski. "Representation-finite biserial algebras", *Journal für die reine und angewandte Mathematik* 345 (1983), pp. 172–181. DOI: 10.1515/crll.1983.345.172. URL: https://doi.org/10.1515/crll.1983.345.172.

[16] Charles A. Weibel. *An Introduction to Homological Algebra*. Vol. 38. Cambridge Studies in Advanced Mathematics. Cambridge: Cambridge University Press, 1994. ISBN: 9780521559874.